A general entropy–tail law for adaptive rare-event estimation

Distributional characterization and certified stopping programs

29 October 2026
Revised after consolidated referee audit

Abstract

A population unit has a hidden positive magnitude drawn from a known law and a hidden sign whose conditional probability may vary arbitrarily with that magnitude, subject to a global rarity constraint. Repeated observations are Bernoulli, and every observation is charged. We characterize the minimax root-mean-square error under a uniform expected budget, allowing arbitrary creation, revisiting, interleaving, and stopping. Up to universal constants, the risk is an entropy–tail selection functional: a selected part of the law pays its rare-event inverse-square scale entropy, and the remainder pays a truncated inverse-gap cost. The result holds jointly for every magnitude law, every allowed rarity, and every positive budget without moment or regularity assumptions. The upper bound uses an observable classifier race; the lower bound combines optimal selection, entropy tiers, scale-sensitive annular saturation, and additive localization. We also derive an exact attained four-prior dual of the robust stopping-flow program, a factor-sixteen comparison with minimax risk, and convergent latent-reveal brackets that certify a sufficient finite cap. An odd-moment argument identifies the fixed-cap bias floor. All specialized proof inputs are included, with their methodological antecedents credited. Exact rational certificates accompany eight finite numerical instances.

Review status. This revision incorporates the consolidated referee audit, including explicit genie-convergence scope, strengthened ordinary-mass and sign-entropy bounds, and clarifications of the adaptive proof interfaces. Its finite numerical bounds have exact rational certificates. External mathematical refereeing and proof-assistant verification of the general theorems remain outstanding. The comparison with prior work is explicit and targeted, without a publication-priority certification.

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1 Model and main results

1.1 The observable experiment

Let μ be a known probability measure on $(0, 1/2]$. Each fresh unit independently samples a hidden magnitude $D \sim \mu$ and a hidden sign $S \mid D = d \sim \mathrm{Ber}(q(d))$. Conditional on (D, S), its observations are iid Bernoulli with mean $1/2 + (2S-1)D$. Different units have independent latent states and observation tapes. The unknown measurable allocation belongs to

$$\mathcal{Q}_\rho = \{q \in L^\infty(\mu) : 0 \le q \le 1,\ \theta(q) := \textstyle\int q\, d\mu \le \rho\}, \qquad 0 < \rho \le 1/4.$$

The target is the positive-sign proportion $\theta(q)$.

An algorithm knows μ, ρ, B, but observes neither D nor S. Using an independent random seed and the observed history, it may request an observation from a fresh or previously observed unit, interleave units, stop, and return a measurable real estimate. Every requested observation costs one, including the first on a unit. Units created but never observed have no statistical effect and may be sampled only when their first observation is requested. If T_A is total work, define

$$R_{\mu,\rho}(B) = \inf_{A:\ \sup_{q\in\mathcal{Q}_\rho} \mathbb{E}_q T_A \le B}\ \sup_{q\in\mathcal{Q}_\rho} \{\mathbb{E}_q(\widehat{\theta}_A - \theta(q))^2\}^{1/2}, \qquad B > 0. \tag{1}$$

This is a uniform expected-budget criterion. Stopping may be unbounded; finite expected work gives almost-sure finite termination under every admissible population. Estimates may be clipped to $[0, \rho]$ without increasing risk. In particular, the constant $\rho/2$ has risk at most $\rho/2$, and randomization permits meaningful procedures even for $B < 1$.

1.2 An entropy–tail selection law

For a submeasure $\sigma \le \mu$, let Γ_σ be its homogeneous dyadic inverse-square cost entropy. With bands $I_j = (2^{-j-2}, 2^{-j-1}]$, put $a_j = \int_{I_j} d^{-2}\, \sigma(dd)$ and $A = \sum_j a_j$. When $0 < A < \infty$,

$$\Gamma_\sigma = A[1 + \mathrm{Ent}((a_j/A)_j)].$$

The finite-fragment definition in Section 3 also covers infinite costs and entropies; $\Gamma_0 = 0$. All logarithms are natural. Our main result is the following uniform comparison:

$$\boxed{R_{\mu,\rho}(B) \asymp \mathcal{K}_{\mu,\rho}(B),} \tag{2}$$

where

$$\mathcal{K}_{\mu,\rho}(B) = \min\left\{\rho,\ \inf_{E \text{ measurable}}\left[\sqrt{\frac{\rho\Gamma_{\mu|E}}{B}} + \int_{E^c} \min\left\{1, \frac{1}{d\sqrt{B}}\right\}\mu(dd)\right]\right\}. \tag{3}$$

Theorem 3.2 states the result formally. Both comparison constants are universal, simultaneously for every probability law on $(0, 1/2]$, every $0 < \rho \le 1/4$, and every $B > 0$. No density, regular variation, positive lower support bound, finite inverse moment, or finite full-law entropy is assumed.

The selected set pays rare-event cost entropy. Its complement pays an ordinary inverse-gap cost for $d > B^{-1/2}$ and only its probability mass below that scale. Equivalently, one minimizes

$$\zeta((0, 1/2]) + \frac{\sqrt{\rho\Gamma_\sigma} + \int d^{-1}\tau(dd)}{\sqrt{B}} \quad \text{over } \sigma + \tau + \zeta = \mu,$$

and caps by ρ. Hard measurable assignments suffice; the optimizer need not be an interval. The split is analytical: the estimator never observes a unit's membership in a selected component.

1.3 Proof structure

The upper construction races a likelihood bank against an ordinary capped classifier on independent observation streams of the same latent unit. Its null likelihood uses the full law μ. A mixed-work bound and finite Poisson multilevel allocation give the three-piece upper bound in Section 3.

The lower proof in Section 4 begins with an optimal finite-band selection. Its ordinary and unresolved pieces are controlled by a variational null-work modulus. Optimality constrains the gaps of the retained bands. Grouping comparable costs and trimming a controlled number of the smallest gaps produces annular components with admissible conditional rarities and active mass-saturation constraints. Allocating rarity proportionally to retained entropy makes their lower bounds add at every budget. Finite-prefix approximation covers laws accumulating at zero.

All specialized ingredients used in these arguments are proved in this manuscript. The appendices include the ordinary classifier, weighted likelihood bank, homogeneous work sum, multilevel allocation, weighted sign entropy, scale-sensitive gate, null-work transfer, and additive localization. Standard concentration, minimax, and convex-duality tools are cited where used. Section 2 distinguishes these ingredients and their antecedents from the contribution of the joint selection law.

1.4 Quantitative stopping certificates

A complementary representation retains more quantitative information. The robust cap-N stopping-flow program $P_N(B)$ has exact attained dual

$$P_N(B)^2 = \max_{q_L,q_H,q_v,q_w\in\mathcal{Q}_\rho} \frac{[\theta(q_H)-\theta(q_L)]_+^2}{4+BK_N(q_L,q_H;q_v,q_w)}.$$

Here K_N is optimal stopped signed-likelihood information relative to q_v per expected observation under q_w. The four priors have separate endpoint, variance, and work roles. Theorem E.1 gives the identity and Proposition E.2 gives its Bellman recursion.

Reference absorption and affinity testing give the explicit comparison

$$R_{\mu,\rho}(B) \le P_{N_*}(B) \le 16R_{\mu,\rho}(B), \qquad N_* = \left\lceil 64\max\{B,\rho B^2\}\right\rceil. \tag{4}$$

The factor sixteen is a program guarantee, distinct from the conservative constants in (2).

A strengthened lower experiment reveals a unit's latent state at its cap. Its dual value G_N obeys

$$0 < G_N \uparrow P_\infty, \qquad P_N \downarrow P_\infty, \qquad G_N/8 \le R_{\mu,\rho}(B) \le P_N.$$

Thus a verified feasible upper policy U and lower dual value L with $U \le 2L$ certify a factor-sixteen approximation. For every fixed input, sufficiently accurate program solutions eventually meet that criterion. No uniform convergence rate or bit-complexity claim is made. Finite-support inputs give genuine finite conic programs. The eight rational examples in Appendix I pass at caps 1, 4, or 8; independent exact checkers verify their feasibility.

The fixed-cap program also has an exact identification threshold: for m distinct positive magnitudes its infinite-budget bias floor vanishes exactly when $N \ge 2m-1$. Infinite-support laws have a positive floor at every fixed finite cap (Theorem H.1). This is compatible with the unrestricted problem, whose cap may increase.

1.5 Reading guide

The main distributional proof is in Sections 3–4. The supporting material is organized as follows.

Proof component	Location
Observable classifiers and multilevel allocation	Appendix A
Weighted sign testing, scale gate, annular saturation	Appendix B
Common hard pairs, adaptive work accounting, additive localization	Appendix C
Stopping-flow representation and observable realization	Appendix D
Exact robust dual and adaptive comparison	Appendices E–F
Convergent finite certificates and fixed-cap identification	Appendices G–H
Exact numerical certificates and reproduction	Appendix I

2 Related work and the scope of the contribution

Adaptive mixtures of coins. Lee–Valiant is the closest direct antecedent [3]. In the extended version, Sections 4–5 develop triangular single-coin estimators and single-coin-to-population lower-bound decompositions. Section 7, Theorems 4–5 and Lemma 22, optimizes an unbiased estimator when both sign-conditional bias distributions, a nominal mixture weight, and a per-coin cap are supplied. Its linear program optimizes information per expected sample. Section 5.2 also reveals a coin's true bias after a cap in a strengthened experiment.

Knowing our marginal magnitude law does not supply those conditional distributions. For $0 < \theta(q) < 1$, the corresponding magnitude laws are

$$\mu_q^+(dd) = \frac{q(d)}{\theta(q)}\,\mu(dd), \qquad \mu_q^-(dd) = \frac{1-q(d)}{1-\theta(q)}\,\mu(dd).$$

They change with the unknown function q. Our optimization must use one observable procedure whose bias, moment, and expected work bounds hold throughout $\mathcal{Q}_\rho$. The exact dual therefore retains separate endpoint, moment-reference, and work-reference populations. The additional statements concern that robust program, its comparison with the full population experiment, and convergent upper and lower certificates. Triangular flows, information/work optimization, adaptive decomposition, and latent revelation are credited to their antecedents.

Gap entropy and sign testing. Chen–Li introduced entropy of normalized dyadic inverse-square gap costs [4, Definition 3.3]. Chen–Li–Qiao proved the weighted Sign-ξ entropy inequality [5, Lemma 4.1]; Appendix D develops its stopping-window argument, and Remark D.4 explicitly covers Bernoulli families away from their boundary. Our weighted sign-testing lemma is an adaptation of this argument. In particular, the form $A[1 + \mathrm{Ent}(\pi)]$ and the statistical origin of its entropy factor are established ideas.

The additional issue here is how much of a known magnitude law should pay that entropy cost. The optimizing set in (6) depends jointly on the law, rarity, and budget. Its complement pays a truncated inverse-gap tail cost. The upper construction implements this decomposition without observing component membership; the lower construction respects both rarity and mass saturation while combining disjoint components. These steps are needed to obtain the all-law selection formula from the sign-testing ingredient. Best-arm identification and latent population-proportion estimation have different targets and sampling access.

Convex estimation, testing, and stopping. Juditsky–Nemirovski give constant-factor affine estimation under their good-pair assumptions, with a Hellinger-affinity characterization [6, Theorem 3.1 and Proposition 3.1]. Polyanskiy–Wu characterize quadratic minimax risk for an affine functional over a convex prior class through a χ^2 modulus for a fixed iid observation kernel [7, Theorem 1]. These results provide the relevant estimation–testing duality. For a fixed capped one-unit policy, its transcript is a fixed observation kernel. Optimizing that policy while controlling expected work uniformly over q, and then comparing with arbitrarily interleaved population algorithms, requires the additional arguments given here. Strong convex duality and occupation-measure formulations of optimal stopping are standard tools [8, 9]; the four-prior formula is a specific application, not a new general duality principle.

Baseline and narrower claims. The supplied manuscript and pinned version-5 development already contain the balanced-law benchmark, rare-event phase results, the variational Ψ lower bound, and the critical quadratic-law entropy correction [1, 2]. The pinned critical proof also credits Chen–Li–Qiao for its sign-entropy mechanism. Those results remain baseline contributions. The specialized ingredients needed here are proved within this manuscript, with their attribution preserved.

The principal added claim is the joint, arbitrary-law entropy–tail characterization, together with a robust stopping-program formulation that admits convergent certificates. The fixed-cap odd-moment identification result is a secondary consequence of this model and elementary moment interpolation; it is not presented as a general mixture-identification principle. The targeted primary-source comparison above identifies the closest checked statements and their hypotheses. It does not certify publication priority.

3 Entropy versus a truncated inverse-gap tail

This section defines a scalar characterization that separates the part of the law handled by rare-event scale entropy from a part handled by the ordinary inverse-gap cost and a part left unresolved at the requested accuracy. We prove its observable upper bound here and its matching lower bound in Section 4. The decomposition is analytical: the algorithm never observes a coin's component label.

3.1 A homogeneous entropy and a selection functional

For a finite submeasure $\sigma \le \mu$, use the canonical bands $I_j = (b_j/2, b_j]$, $b_j = 2^{-(j+1)}$, and put

$$a_j(\sigma) = \int_{I_j} d^{-2}\,\sigma(dd), \qquad A_\sigma = \sum_j a_j(\sigma).$$

For a finite nonnegative vector a, define

$$\mathcal{G}(a) = A + \sum_j a_j \log(A/a_j), \qquad A = \sum_j a_j, \qquad \mathcal{G}(0) = 0.$$

Zero coordinates contribute zero. The functional $\mathcal{G}$ is concave, coordinatewise increasing, homogeneous of degree one, and superadditive. Indeed, on positive coordinates its derivative is $1 + \log(A/a_j)$ and its Hessian is $A^{-1}\mathbf{1}\mathbf{1}^\top - \operatorname{diag}(a_j^{-1})$, which is negative semidefinite by Cauchy–Schwarz. Continuity gives the boundary cases, and concavity plus homogeneity gives superadditivity. Define

$$\Gamma_\sigma = \sup_J \mathcal{G}(a_0(\sigma), \dots, a_J(\sigma)). \tag{5}$$

Thus $\Gamma_0 = 0$, and, when $0 < A_\sigma < \infty$, $\Gamma_\sigma = A_\sigma[1 + \mathrm{Ent}((a_j/A_\sigma)_j)]$, with extended-value interpretation. Also write $H_\tau = \int d^{-1}\,\tau(dd)$ for any finite submeasure τ.

Let $h = B^{-1/2}$ and define

$$\boxed{\mathcal{K}_{\mu,\rho}(B) = \min\left\{\rho, \inf_{0\le\sigma\le\mu}\left[h\sqrt{\rho\Gamma_\sigma} + \int \min\{1, h/d\}\, d(\mu-\sigma)\right]\right\}.} \tag{6}$$

Equivalently,

$$\mathcal{K}_{\mu,\rho}(B) = \min\left\{\rho, \inf_{\sigma+\tau+\zeta=\mu}\left[\zeta((0,1/2]) + \frac{\sqrt{\rho\Gamma_\sigma} + H_\tau}{\sqrt{B}}\right]\right\}. \tag{7}$$

Indeed, at fixed σ, assign each remaining magnitude to the cheaper of the costs 1 and h/d.

Lemma 3.1 (Hard assignments suffice)**.** *The infimum in (6) is unchanged if $\sigma = \mu|S$ for a measurable set S. In particular, for a finite atomic law a minimizing assignment can put each atom entirely in one of the three pieces in (7).*

Proof. Write $f = d\sigma/d\mu \in [0,1]$ and $S_t = \{f > t\}$, so $\sigma = \int_0^1 \mu|S_t\, dt$. Concavity of the square root of every finite-fragment entropy gives

$$\sqrt{\Gamma_\sigma} \ge \int_0^1 \sqrt{\Gamma_{\mu|S_t}}\, dt$$

after monotone convergence in the fragment size. For a finite-objective input the relevant entropies are finite for $t > 0$, since $\mu|S_t \le \sigma/t$ and $\Gamma_{\mu|S_t} \le \Gamma_\sigma/t$. The remainder integral is linear, so the objective at σ is at least the average of the hard-set objectives. Their infimum is therefore no larger. The reverse inequality is immediate. For a finite law the minimum is attained among the finitely many assignments. This is a selection problem, not a claim of convex minimization over its assignments. □

The definition immediately implies

$$\mathcal{K}_{\mu,\rho}(B) \le \min\{\rho, \omega_\mu(B^{-1/2}), \sqrt{\rho\Gamma_\mu/B}\}, \qquad \omega_\mu(h) = \int \min\{1, h/d\}\, \mu(dd).$$

Infinite quantities are allowed in this display.

Theorem 3.2 (An entropy–tail law for the full joint problem)**.** *There are universal constants $0 < c < C < \infty$ such that, for every probability measure μ on $(0, 1/2]$, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$\boxed{c\,\mathcal{K}_{\mu,\rho}(B) \le R_{\mu,\rho}(B) \le C\,\mathcal{K}_{\mu,\rho}(B).} \tag{8}$$

The functional has the equivalent selection and three-piece forms (6)–(7), and hard measurable assignments suffice. No density, regular variation, finite inverse moment, or finite full-law entropy is required.

The upper half is Theorem 3.3; the lower half is Theorem 4.1. The comparison constants in (8) are universal but conservative. They are not the factor sixteen proved for the finite convex program; these are different representations and different comparison bounds. The proof also gives the explicit conservative constants in (41).

Theorem 3.3 (Three-piece observable upper bound)**.** *There is a universal constant C such that, for every magnitude law on $(0, 1/2]$, every $0 < \rho \le 1/4$, and every $B > 0$,*

$$R_{\mu,\rho}(B) \le C\mathcal{K}_{\mu,\rho}(B). \tag{9}$$

No component label or magnitude is revealed to the algorithm.

3.2 An uncapped reference bank with a capped observable race

Fix a decomposition $\mu = \sigma + \tau + \zeta$, put $z = \zeta((0, 1/2])$, and assume $\Gamma_\sigma, H_\tau < \infty$. It suffices to treat $z \le \rho/128$: otherwise the constant $\rho/2$ has risk at most $64z$. Write $A = A_\sigma$, $H = H_\tau$. If $\sigma \neq 0$, use all its occupied bands, with weights $\pi_j = a_j(\sigma)/A$ and references $b_j/2$ and $b_j/4$. For an internal allowance $u < \rho$, set $\epsilon = u/\rho$. Stop negative if a negative reference likelihood against fair reaches $1/(\epsilon\pi_j)$; stop positive if

$$\frac{\sum_j \pi_j P_{1/2+b_j/4,n}}{\int P_{1/2-d,n}\,\mu(dd)} \ge 1/u. \tag{10}$$

Give negative crossings priority. A zero denominator triggers a positive stop. The split $\sigma+\tau+\zeta = \mu$ specifies the references and the work analysis; membership in these submeasures is never observed. The known all-negative mixture in the denominator is the full law μ, regardless of the decomposition.

By Lemma A.3, a wrong negative crossing has conditional positive-sign probability at most ϵ, and a positive crossing has all-negative-mixture probability at most u. Thus the total probability of a wrong bank stop is at most $2u$ over $\mathcal{Q}_\rho$. Termination outside σ is not assumed.

Race this bank against the ordinary classifier of Lemma A.1, using alternating observations of the same physical coin. At an ordinary cap M, its conditional work certificate is

$$o_M(d) = \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}.$$

For any finite measure ξ, put

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\,\xi(dd), \qquad O_\xi(M) = \int o_M(d)\,\xi(dd).$$

The ordinary error under μ is at most $\eta_\mu(M)$. Stop as soon as either component stops and return that output. On completed counterfactual streams, the physical work is at most

$$2\min\{T_{\text{bank}}, T_{\text{ordinary}}\} \le 2M.$$

A wrong output implies either a wrong bank stop or an ordinary error. When $\sigma = 0$, omit the bank and use the ordinary classifier alone.

On a σ-almost-every gap d in band j, the matching negative reference has drift at least $4d^2/3$ and log overshoot below one. Its expected crossing time is at most

$$\frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{11}$$

On a positive gap, the event $\widehat{p}_n - 1/2 \ge d/2$ gives

$$\frac{P_{-,\mu,n}}{P_{\text{f},n}} \le \int e^{-2nx^2}\,\mu(dx) \le z + \frac{A}{2en} + \frac{H}{2\sqrt{en}}. \tag{12}$$

The last two bounds use, respectively, $\sup_x x^2 e^{-2nx^2} = 1/(2en)$ and $\sup_x x e^{-2nx^2} = 1/(2\sqrt{en})$. For $n \ge d^{-2}$ the denominator is at most $z + Ad^2 + Hd$, while the matching positive reference contributes at least $\pi_j e^{nd^2/3}$ relative to fair. Equation (46) therefore bounds positive expected bank work by

$$\frac{6}{d^2}\left[1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+ \frac{z + Ad^2 + Hd}{\rho}\right]. \tag{13}$$

These arguments prove finite conditional bank work on σ. On $\tau+\zeta$, the ordinary cap guarantees termination of the race.

Since $z/\rho \le 1$,

$$\log_+ \frac{z + Ad^2 + Hd}{\rho} \le \log 3 + \log_+(Ad^2/\rho) + \log_+(Hd/\rho).$$

Using $\sup_{x\ge 1} \log x/x = 1/e$ and $\sup_{x\ge 1} \log x/x^2 = 1/(2e)$ gives

$$\int q(d)d^{-2} \log_+ \frac{z + Ad^2 + Hd}{\rho}\, \sigma(dd) \le (\log 3 + 1/e)A + \frac{H^2}{2e\rho}. \tag{14}$$

Here $\int q\, d\sigma \le \rho$ and $q \le 1$ follow from the original global rarity class; no separate conditional rarity assumption is used. Integrating (11) and (13) therefore bounds the bank work on σ by

$$6\Gamma_\sigma + 6A \log(1/\epsilon) + 6(\log 3 + 1/e)A + \frac{3H^2}{e\rho}. \tag{15}$$

3.3 A resolved-law cap schedule

Let $\nu = \sigma + \tau$ and set

$$\delta_\ell = \rho 16^{-(\ell+1)}, \qquad u_\ell = \delta_\ell/4, \qquad \epsilon_\ell = u_\ell/\rho.$$

Choose M_ℓ as the least power of four, at least four, such that $\eta_\nu(M_\ell) \le \delta_\ell/8$. Every chosen cap has a predecessor because $\eta_\nu(4) = 1 - z$. The inequality $\eta_\nu(4M) \ge \eta_\nu(M)/16$ implies

$$\delta_\ell/128 < \eta_\nu(M_\ell) \le \delta_\ell/8. \tag{16}$$

Thus the selected powers of four are distinct and increasing.

If $z > 0$, let J be the largest integer with $\delta_J \ge 8z$. Then

$$8z \le \delta_J < 128z. \tag{17}$$

For $\ell \le J$, $\eta_\mu(M_\ell) \le \eta_\nu(M_\ell) + z \le \delta_\ell/4 = u_\ell$. The raced classifier has total error at most $3u_\ell \le \delta_\ell$ and false-positive probability at most $2u_\ell \le \delta_\ell$. When $z = 0$, these conclusions hold at every level.

Since $\log(1/\epsilon_\ell) = \log 64 + \ell \log 16$ and $A \le \Gamma_\sigma$, (15) is at most $40(\ell+1)\Gamma_\sigma + 2H^2/\rho$. A uniform raced-work certificate is consequently

$$C_\ell = 80(\ell+1)\Gamma_\sigma + 4H^2/\rho + 2O_\tau(M_\ell) + 2zM_\ell. \tag{18}$$

It remains a valid, possibly loose, bound when $\sigma = 0$ and the bank is omitted.

The binary level-zero output has second moment at most $\rho + \delta_0 = 17\delta_0$. An adjacent difference, run on successive unused blocks of the same coin, has second moment at most $\delta_\ell + \delta_{\ell-1} = 17\delta_\ell$. Set $v_\ell = 17\delta_\ell$, $k_0 = C_0$, $k_\ell = C_\ell + C_{\ell-1}$ for $\ell \ge 1$, and $S_J = \sum_{\ell\le J} \sqrt{v_\ell k_\ell}$. Independent-Poisson allocation across these corrections (Lemma A.5) gives RMS at most

$$\sqrt{\delta_J^2 + S_J^2/B}. \tag{19}$$

The seed may choose zero replications, and the Poisson means may be below one. Every first and subsequent observation is charged. Every such finite-level procedure has finite deterministic local caps. The reference bank may be countable; its stopping events are measurable countable unions and its positive reference is a probability mixture.

3.4 Controlling mixed work without an extra logarithm

For every finite τ, the following bounds hold:

$$O_\tau(M) \le 21H\sqrt{M}, \qquad \eta_\sigma(M) \le A/M, \qquad \eta_\tau(M) \le H/\sqrt{M}. \tag{20}$$

The latter two are pointwise inequalities before integration. For the first, put $x = \sqrt{M}\,d$. The ratio of $o_M(d)$ to $\sqrt{M}/d$ is at most one for $x \le 1$, and at most $17(1 + 2\log x)/x \le 34/\sqrt{e} < 21$ for $x \ge 1$. Lemma A.2 supplies the homogeneous ordinary kernel bound:

$$\sum_{k\ge 1} \sqrt{\eta_\tau(4^k) O_\tau(4^k)} \le 208H. \tag{21}$$

Its proof applies without normalization of τ: dyadic binning yields the summable kernels 2^{k-j} for $j \ge k$, and $2^{3(j-k)}$, $2^{j-k}(1 + k - j)$ for $j < k$. Apply $\sqrt{xy} \le (x + y)/2$ and sum these kernels.

Split levels according to which term of $\eta_\nu = \eta_\sigma + \eta_\tau$ is at least half the total. On levels where η_τ dominates, $\delta_\ell \le 256\eta_\tau(M_\ell)$, so distinctness of the caps and (21) give a total $\sum \sqrt{\delta_\ell O_\tau(M_\ell)} \le 3328H$. On the other levels,

$$\delta_\ell \le 256\eta_\sigma(M_\ell) \le 256A/M_\ell, \qquad M_\ell \le 256A/\delta_\ell.$$

Therefore

$$\sqrt{\delta_\ell O_\tau(M_\ell)} \le 4\sqrt{21}\, H^{1/2} A^{1/4} \delta_\ell^{1/4}.$$

The sum of $\delta_\ell^{1/4}$ over all levels is $\rho^{1/4}$. The arithmetic–geometric mean inequality proves

$$\sum_{\ell\le J} \sqrt{\delta_\ell O_\tau(M_\ell)} \le C_{\mathrm{mix}}[H + \sqrt{\rho A}]. \tag{22}$$

The work on the unresolved mass is controlled separately. On levels where η_σ dominates,

$$\sum \sqrt{z\delta_\ell M_\ell} \le 16(J + 1)\sqrt{zA} \le \sqrt{2\rho A}. \tag{23}$$

The last inequality uses $z \le \delta_J/8$ and $(J + 1)4^{-(J+1)} \le 1/4$. On levels where η_τ dominates, $\sqrt{M_\ell} \le 256H/\delta_\ell$, so

$$\sum \sqrt{z\delta_\ell M_\ell} \le 256H\sqrt{z} \sum_{\ell=0}^{J} \delta_\ell^{-1/2} \le \frac{1024}{3} H\sqrt{z/\delta_J} \le \frac{1024}{3\sqrt{8}} H. \tag{24}$$

When $z = 0$ these contributions vanish.

Completion of Theorem 3.3. The shifted adjacent-level sum has a factor $1/4$, so

$$S_J \le \frac{5\sqrt{17}}{4} \sum_{\ell=0}^{J} \sqrt{\delta_\ell C_\ell}.$$

Use (18), (22), (23), (24), and $A \le \Gamma_\sigma$. The elementary sums $\sum_{\ell\ge 0} 4^{-\ell}\sqrt{\ell + 1} \le 16/9$ and $\sum_{\ell\ge 0} 4^{-\ell} = 4/3$ then give

$$S_J \le C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H].$$

For $z > 0$, (17) and (19) imply $R(B) \le 128z + C_{\mathrm{sum}}[\sqrt{\rho\Gamma_\sigma} + H]/\sqrt{B}$. For $z = 0$, the bound on S_J holds at every finite terminal level. Use (19) and then $\delta_J \to 0$; taking the infimum over these finite constructions gives the asserted bound. The constant estimator covers $z > \rho/128$. Finally take the infimum over decompositions and compare with $\rho/2$. All approximations are at fixed (μ, ρ, B). The comparison constant is independent of the chosen decomposition and terminal level; no attaining infinite-dimensional decomposition is required. No infinite-algorithm interchange is required. The same infimum can be approached with finite reference banks: finite Γ_σ implies $H_\sigma < \infty$, so truncate σ to finitely many bands and move its remainder to τ. The entropy does not increase and the additional inverse-gap cost tends to zero. Then use the finite-level construction. □

4 A matching scalar lower bound by entropy packing

The lower proof uses optimality of the retained set in (6). Its ordinary and ignored pieces are controlled by the variational modulus proved in Appendix C. Its retained entropy can be packed into disjoint components on which the actual annular mass constraint is active. The additive localization theorem then combines these components at every budget.

Theorem 4.1 (Matching entropy–tail lower bound)**.** *There is a universal $c > 0$ such that, for every magnitude law, $0 < \rho \le 1/4$, and $B > 0$,*

$$R_{\mu,\rho}(B) \ge c\,\mathcal{K}_{\mu,\rho}(B).$$

Together with Theorem 3.3, this proves Theorem 3.2.

4.1 Canonical endpoints change the functional by at most two

Put $h = B^{-1/2}$, $w_j = \mu(I_j)$, and $a_j = w_j/b_j^2$, where $b_j = 2^{-(j+1)}$ and $I_j = (b_j/2, b_j]$. For a finite prefix, define

$$k_J = \min\left\{\rho,\ \min_{I\subset\{0,\dots,J\}}\left[h\sqrt{\rho\,\mathcal{G}((a_j)_{j\in I})} + \sum_{\substack{j\le J\\ j\notin I}} w_j \min\{1, h/b_j\}\right]\right\}. \tag{25}$$

Fractional retained band masses cannot improve this finite expression: the square root of $\mathcal{G}$ is concave, the remainder is linear, and the minimum over a finite splitting box occurs at a vertex. The values k_J increase to a limit k, with

$$k_J \le k \le \min\{\rho, k_J + \sum_{j>J} w_j\}. \tag{26}$$

Restriction proves the first inequality; assigning every omitted band to ignored mass proves the second. The limit is at fixed (μ, ρ, B); no law-independent truncation depth or nested choice of minimizers is required. The omitted tail probability must be retained before the limit.

The exact and canonical functionals satisfy

$$k \le \mathcal{K}_{\mu,\rho}(B) \le 2k. \tag{27}$$

Indeed, on a band the exact inverse-square cost lies between its canonical cost and four times that cost, while

$$\min\{1, h/b_j\} \le \min\{1, h/d\} \le 2\min\{1, h/b_j\}.$$

Monotonicity and homogeneity of $\mathcal{G}$ give the factor two after taking the square root. The lower comparison applies to any fractional submeasure before minimizing the canonical splitting box. For the upper comparison, lift a canonical choice to a union of whole original bands and assign the vanishing omitted tail to ignored mass. These are comparisons of functionals. We do not replace the original observation experiment by an experiment with rounded magnitudes.

We prove $R_{\mu,\rho}(B) \ge ck_J$ uniformly in the finite prefix. The limit (26) and (27) will complete the theorem.

4.2 Two consequences of optimal finite selection

Fix a minimizer I of the inner finite objective in (25), before its outer cap by ρ. Let O consist of the complementary bands with $b_j \ge h$, and Z of those with $b_j < h$. Write

$$F = \mathcal{G}((a_j)_{j\in I}), \qquad A = \sum_{j\in I} a_j, \qquad z = \sum_{j\in Z} w_j, \qquad H_O = \sum_{j\in O} a_j b_j.$$

The minimized inner cost is $z + hH_O + h\sqrt{\rho F}$. Zero-mass bands can be omitted.

Lemma 4.2 (Optimal-band inequalities)**.** *Every ordinary band $j \in O$ has $w_j \le \rho$. If $F > 0$ and $\lambda = \sqrt{\rho/F}$, every retained band satisfies*

$$b_j \ge \frac{\lambda}{2}[1 + \log(A/a_j)]. \tag{28}$$

Proof. For an ordinary band, put $x = a_j$ and compare with moving it to I. Optimality gives

$$x b_j \le \sqrt{\rho}[\sqrt{F + \Delta} - \sqrt{F}].$$

If $F = 0$, the squared bracket is x. Otherwise use $\log(1 + t) \le \sqrt{t}$ for $t \ge 0$ to obtain

$$\Delta = x + A\log(1 + x/A) + x\log(1 + A/x) \le x + 2\sqrt{Ax} \le x + 2\sqrt{Fx}.$$

Thus $\sqrt{F + \Delta} - \sqrt{F} \le \sqrt{x}$, and optimality gives $x^2 b_j^2 \le \rho x$, proving $w_j \le \rho$.

For $j \in I$, removing the band and assigning it to ordinary work is admissible even if ignoring it would be cheaper. Hence

$$a_j b_j \ge \sqrt{\rho}[\sqrt{F} - \sqrt{F_{-j}}].$$

The derivative of $\mathcal{G}$ in this coordinate is $1 + \log(A/a_j)$; it only increases as that coordinate is removed. Therefore $F - F_{-j} \ge a_j[1 + \log(A/a_j)]$. Dividing by $\sqrt{F} + \sqrt{F_{-j}} \le 2\sqrt{F}$ proves (28). □

4.3 The ordinary and ignored costs are already lower bounds

Recall the functional from Proposition C.4:

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0 \le q \le 1 \\ \int q\, d\mu \le \rho}} \int \min\{q(d), h\sqrt{q(d)/d}\}\, \mu(dd).$$

Theorem C.3 and Proposition C.4 imply

$$R_{\mu,\rho}(B) \ge \Psi_{\mu,\rho}(B^{-1/2})/8192. \tag{29}$$

For an explicit deduction, take $b = \Psi(h)/64 \le \rho/64$ in the bias-indexed null obstruction in Proposition C.4. Evaluating it at $h = B^{-1/2}$ gives $J \ge (15\Psi(h)/16)^2/(192h^2)$. The bias-indexed transfer has its smaller branch at b, yielding (29).

Every ignored band has $d \le b_j < h$, so allocating on these bands gives

$$\min\{\rho, z\} \le \Psi_{\mu,\rho}(h). \tag{30}$$

For the ordinary bands we need an elementary dyadic inequality. If masses v_j satisfy $v_j \le s$ at values $f_j = b_j^{-1} = 2^{j+1}$, then

$$\sum_j v_j f_j \le 2 \sup_{\substack{0 \le q_j \le 1 \\ \sum_j v_j q_j \le s}} \sum_j v_j q_j f_j. \tag{31}$$

If the total mass is at most s, this is immediate. Otherwise select the largest f_j up to mass s, allowing a fraction at the boundary index j_0. Let r be the selected mass at the boundary index. The selected mass above it is $s - r$, at values at least $2f_{j_0}$, so the selected integral is at least $(2s - r)f_{j_0}$. The excluded smaller values contribute at most sf_{j_0}, and the excluded boundary

mass contributes at most $(s-r)f_{j_0}$. Thus the excluded integral is no larger than the selected integral, proving (31).

Apply it to O with $s=\rho$, using Lemma 4.2. It follows that H_O is at most twice the largest ordinary-band inverse-gap integral at mass ρ. On these bands $h/b_j \le 1$ and $d \le b_j$; for every $0 \le q \le 1$, $\min\{q, h\sqrt{q}/d\} \ge qh/b_j$. Consequently

$$hH_O \le 2\Psi_{\mu,\rho}(h). \tag{32}$$

4.4 Packing entropy into comparable-cost groups

Lemma 4.3 (Entropy tiers)**.** *For a finite positive vector a, let $M = \max_j a_j$ and group its coordinates by $a_j \in (M2^{-k-1}, M2^{-k}]$, $k \ge 0$. If F_k is each group's entropy cost, then*

$$\mathcal{G}(a) \le 7\sum_k F_k. \tag{33}$$

Proof. Let A_k be the cost mass in group k, n_k its cardinality, and $p_k = A_k/A$. Entropy decomposition gives $\mathcal{G}(a) = \sum_k F_k + A\,\mathrm{Ent}(p)$. Comparison with the probability vector proportional to $2^{-k/2}$ gives

$$A\,\mathrm{Ent}(p) \le A\log(2+\sqrt{2}) + \frac{\log 2}{2}\sum_k kA_k.$$

On groups with $n_k \ge 2^{k/2}$, the second term is at most $\sum_k A_k \log n_k \le \sum_k F_k$; comparable weights imply $F_k \ge A_k[1+\log n_k - \log 2] \ge A_k \log n_k$. On the other groups, $A_k \le M2^{-k/2}$, so their contribution is at most

$$\frac{\log 2}{2} M \sum_{k\ge 0} k2^{-k/2} < 3M \le 3A.$$

Finally $\log(2+\sqrt{2}) < 2$ and $A \le \sum_k F_k$. Adding the initial $\sum_k F_k$ proves (33). □

Split the retained low-gap bands $j \ge 6$ by index parity and then by these cost tiers. Parity splitting loses at most a factor two in entropy, so all resulting group entropies sum to at least $F_{\mathrm{low}}/14$. Every group is a separated annular family with endpoint ratios at least four, and its cost weights lie in some interval $(c, 2c]$.

Consider a group of n bands and cost entropy at most $2cn(1+\log n)$. Since $A \ge nc$ and every group coordinate is at most $2c$, (28) implies

$$b_j \ge \frac{\lambda}{2}[1+\log(n/2)] \ge \frac{\lambda}{8}(1+\log n).$$

Remove the $t = \lceil \log_4 n \rceil$ smallest-gap bands, with $t=0$ for $n=1$. For $n \ge 2$, $t \le n/2$. The remaining entropy G_i satisfies

$$G_i \ge F_{\mathrm{group}}/16. \tag{34}$$

Indeed, at least $n/2$ coordinates remain, each at least c. Coordinate monotonicity gives entropy at least $cn'(1+\log n')$, whereas the original entropy is at most $2cn(1+\log n)$. Since $1+\log(n/2) \ge (1-\log 2)(1+\log n)$, their ratio is at least $(1-\log 2)/4 > 1/16$. The case $n=1$ is immediate.

Separation amplifies the smallest remaining endpoint by at least $4^t \ge n$. Hence

$$b_{\min,i} \ge \lambda n(1+\log n)/8. \tag{35}$$

Here G_i is the canonical endpoint entropy of the retained bands, which is exactly the annular quantity $\Gamma^\circ_{\sigma_i}$ in Appendix B. Let σ_i be the original law restricted to these remaining bands, τ_i

its mass, a_i its largest endpoint, and $w_{*,i}$ its smallest unnormalized band mass. Then $\tau_i/a_i^2 \geq c$ and $w_{*,i} \geq cb_{\min,i}^2$. Together with $G_i \leq 2cn(1+\log n)$, this gives the actual annular saturation bound

$$\boxed{\frac{\tau_i w_{*,i}}{a_i^2} \geq \frac{\lambda^2 G_i^2}{256}.} \tag{36}$$

This step retains the minimum-band-mass constraint; it does not replace that constraint by an inverse-gap moment.

Allocate original rarity and conditional rarity by

$$\theta_i = \frac{\lambda^2 G_i}{1024}, \qquad r_i = \theta_i/\tau_i. \tag{37}$$

Equation (36) makes $\theta_i G_i$ at most one quarter of the saturation term. Since $G_i \geq \tau_i/a_i^2$, that saturation divided by G_i is at most $w_{*,i} \leq \tau_i$. Thus $\theta_i \leq \tau_i/4$ and $r_i \leq 1/4$. The components are disjoint, and superadditivity gives

$$\sum_i \theta_i \leq \lambda^2 F/1024 = \rho/1024.$$

Every retained component is supported on $(0, 1/128]$; parity gives endpoint ratios at least four. The normalized minimum band mass is $w_{*,i}/\tau_i$, so (36) is exactly the unnormalized saturation condition required by the annular proposition. Every component is therefore admissible, and every entropy branch of the scale-uniform annular lower bound is active.

4.5 Combining the components at every budget

Let F_{high} be the retained entropy in the first six bands and F_{low} the remainder. Entropy decomposition gives $F \leq 2(F_{\text{high}} + F_{\text{low}})$, and $F_{\text{high}} \leq 4096(1+\log 6)$. If $F_{\text{high}} \geq F_{\text{low}}$, then F is bounded by an absolute constant. The feasible constant allocation $q = \rho$ gives $\Psi(h) \geq \min\{\rho, 2h\sqrt{\rho}\}$, so

$$\min\{\rho, h\sqrt{\rho F}\} \leq C\Psi(h). \tag{38}$$

Otherwise $F_{\text{low}} > F_{\text{high}}$ and $F \leq 4F_{\text{low}}$. Lemma 4.3, parity splitting, and (34) give

$$\sum_i G_i \geq F_{\text{low}}/224 \geq F/896. \tag{39}$$

Proposition B.4 and Theorem C.7 now imply

$$R_{\mu,\rho}(B) \gtrsim \sum_i \min\{\theta_i, \sqrt{\theta_i G_i/B}\}.$$

All conditional bias and annular constants here are fixed universal constants. Crucially, the ratio $\theta_i/G_i = \lambda^2/1024$ is common to every component. This remains true with the fixed conditional-bias and annular constants retained, so all components have the same budget crossover. Therefore the displayed sum equals

$$\left(\sum_i G_i\right) \min\{\lambda^2/1024, \lambda/(32\sqrt{B})\}.$$

Using (39) and $\lambda^2 = \rho/F$ proves

$$R_{\mu,\rho}(B) \geq c \min\{\rho, h\sqrt{\rho F}\}. \tag{40}$$

No limit in B or separate lower threshold for each component is used.

Completion of Theorem 4.1. If $z \geq \rho$, (30) and (29) already give $R \geq ck_J$. Otherwise (30) and (32) imply

$$k_J \leq \min\{\rho, 3\Psi(h) + h\sqrt{\rho F}\} \leq 3\Psi(h) + \min\{\rho, h\sqrt{\rho F}\}.$$

Apply (29) and either (38) or (40). If $F = 0$, the Ψ bound alone suffices. Thus $R_{\mu,\rho}(B) \geq ck_J$ with one universal constant for every finite prefix. Let J increase and use (26)–(27). This proves the theorem. The passage to every magnitude law uses only vanishing omitted tail probability and finite-band lower constructions. □

4.6 Explicit conservative comparison constants

The preceding arguments provide, without optimizing constants,

$$10^{-17}\mathcal{K}_{\mu,\rho}(B) \leq R_{\mu,\rho}(B) \leq 33000\mathcal{K}_{\mu,\rho}(B). \tag{41}$$

For completeness, the work summation in Section 3 has entropy and ordinary coefficients at most, respectively,

$$\frac{5\sqrt{17}}{4}\left(\frac{4\sqrt{80}}{9} + 2\sqrt{42} + 2\right) < 104,$$

$$\frac{5\sqrt{17}}{4}\left(\frac{2}{3} + \sqrt{2}(3328 + 2\sqrt{21}) + \frac{512}{3}\right) < 27000.$$

Together with the $128z$ bias term and the constant estimator, these give the rounded upper constant.

For the lower bound let $c_{\mathrm{ann}} = e^{-28}/2000000$ and $c_E = \sqrt{c_{\mathrm{ann}}}/(896 \cdot 128 \cdot 32)$. Before suppressing constants, the packed contribution is at least

$$\frac{\sum_i G_i}{128} \min\left\{\frac{\lambda^2}{65536}, \frac{\sqrt{c_{\mathrm{ann}}}\lambda}{32\sqrt{B}}\right\} \geq c_E \min\{\rho, h\sqrt{\rho F}\}.$$

The high-gap alternative gives $\min\{\rho, h\sqrt{\rho F}\} \leq 112\Psi(h)$ and has a stronger constant. Combining (29) with the final $3\Psi(h)$ assembly and $\mathcal{K} \leq 2k$ gives the coefficient $[2(24576 + c_E^{-1})]^{-1} > 10^{-17}$. For example, $e < 3$ and $\sqrt{2000000} < 1500$ imply $2(24576 + 896 \cdot 128 \cdot 32 \cdot 1500 \cdot 3^{14}) < 10^{17}$. These are constants for the scalar comparison, not the factor sixteen of the stopping-program certificate.

5 Consequences and limits of the characterization

The selection formula recovers two immediate upper bounds by choosing the retained set to be all or none of the magnitude space:

$$R_{\mu,\rho}(B) \lesssim \min\left\{\rho, \sqrt{\rho\Gamma_\mu/B}, \int \min\{1, (d\sqrt{B})^{-1}\}\,\mu(dd)\right\}.$$

The full formula additionally allows different regions of the same law to use different treatments. Its lower proof explains why contributions from several irregular components may need to be combined.

For a single magnitude $\mu = \delta_d$, hard assignments leave two choices and give

$$R_{\delta_d,\rho}(B) \asymp \min\{\rho, \sqrt{\rho/(Bd^2)}\}.$$

For a general law, the finite-prefix selection values k_J in (25) have an explicit tail-mass error bound (26), and their limit lies between $\mathcal{K}/2$ and $\mathcal{K}$. This is a finite approximation using only

known band masses. Exhaustive selection is possible, but an efficient general selection algorithm has not been established here.

The scalar characterization concerns the full minimax risk, whereas P_N bounds worst bias and worst variance separately. Accordingly, its exact dual is an identity for a certificate program, and the statistical connection is a comparison inequality. The convergent genie brackets can certify that a chosen cap suffices for a particular input without using the large universal cap. They do not give a uniform runtime bound.

The remaining quantitative questions include sharp comparison constants, efficient selection for structured laws, and rates of cap convergence. The present theorem assumes known μ, Bernoulli observations, and a uniform expected observation budget. Extensions to an unknown magnitude law, other observation kernels, or almost-sure budgets require separate arguments.

This consolidated manuscript includes the specialized proof chain rather than relying on the earlier working notes. Separate model audits found no remaining substantive inconsistency in the reviewed chain; exact arithmetic verifies the finite numerical certificates. These checks do not constitute external refereeing or proof-assistant verification of the general theorems. The related-work comparison is targeted and does not certify publication priority.

5.1 Two scope examples

A noninterval optimal assignment. Let $\varepsilon = 1/1024$, put masses $(1-\varepsilon)/2$, ε, and $(1-\varepsilon)/2$ at $1/2$, $1/4$, and $1/8$, respectively, and take $\rho = 1/4$, $h = B^{-1/2} = 1/1024$. All three atoms lie above h. Enumerating the eight hard retained sets shows that the unique minimum retains $\{1/2, 1/8\}$ and treats $1/4$ ordinarily. The following rational enclosures for the inner cost divided by h make the strict comparison explicit:

Retained atoms	Enclosure for cost divided by h
$\varnothing$	$(4.99902, 4.99903)$
$\{1/2\}$	$(4.70676, 4.70677)$
$\{1/4\}$	$(5.05761, 5.05762)$
$\{1/2, 1/4\}$	$(4.72155, 4.72156)$
$\{1/8\}$	$(3.82997, 3.82998)$
$\{1/2, 1/8\}$	$(3.22748, 3.22749)$
$\{1/4, 1/8\}$	$(3.83271, 3.83272)$
$\{1/2, 1/4, 1/8\}$	$(3.22943, 3.22944)$

These follow by substituting the rational weights into $\mathcal{G}$. For example, after reducing each logarithm argument to $1 \le x \le 2$, use $y = (x-1)/(x+1)$ and

$$0 \le \log x - 2\sum_{k=0}^{m-1} \frac{y^{2k+1}}{2k+1} \le \frac{2y^{2m+1}}{(2m+1)(1-y^2)},$$

together with rational bounds for $\log 2$ from the same series and square-root bounds checked by squaring. Already $m = 90$ gives intervals much narrower than those displayed. The companion verifier `checks/verify_revisions.py` performs these exact calculations. Since the outer cap is inactive, the original functional has the same optimizer. No interval can contain both retained atoms while excluding the intervening positive-mass atom.

Infinite inverse moments and a slow tail. Let $\mu\{2^{-j}\} = 1/[j(j+1)]$ for $j \ge 1$. Both inverse moments of orders one and two are infinite. At $B = 4^J$, ignore $j > J$ and treat the first

J atoms ordinarily. This gives

$$\mathcal{K}(B) \le \frac{1}{J+1} + \sum_{j=1}^{J} \frac{2^{j-J}}{j(j+1)} = \frac{1}{J+1} + O(J^{-2}).$$

For fixed ρ and sufficiently large J, the ignored-tail allocation gives $\Psi(2^{-J}) \ge 1/(J+1)$. Thus the upper and lower scales match even though the full-law inverse moments diverge. This example uses only the vanishing tail probability, not an integrable full-law cost.

A Observable classification and finite Poisson allocation

This section proves the specialized upper-bound tools used below. The ordinary classifier is the construction in the supplied manuscript [1, 2]; the weighted reference bank and multilevel allocation are restated with their proofs so that the present paper does not depend on the preceding unpublished notes. All likelihoods refer to the actual observed binary prefix. A reference distribution is used to evaluate that prefix, and never reveals or redraw the latent state of a unit.

A.1 A capped classifier that does not know the gap

For a dyadic integer $M \geq 1$, inspect prefix lengths $1, 2, 4, \ldots, M$. At a prefix $n < M$, stop if

$$\left|\widehat{p}_n - \frac{1}{2}\right| > t_{n,M} := \sqrt{\frac{1 + \log(M/n)}{n}},$$

and output the side of $1/2$ containing $\widehat{p}_n$. At M, output $\mathbf{1}_{\{\widehat{p}_M > 1/2\}}$, with a negative output on ties. Denote the output and work by $(Z_M^{\mathrm{ord}}, T_M^{\mathrm{ord}})$.

Lemma A.1 (Ordinary conditional error and work)**.** *For every $0 < d \leq 1/2$ and either latent sign,*

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq \min\{1, (Md^2)^{-2}\}, \tag{42}$$

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq o_M(d) := \min\{M, 17d^{-2}[1 + \log_+(Md^2)]\}. \tag{43}$$

Proof. For independent Bernoulli observations of mean p, Hoeffding's inequality gives $\mathbb{P}(\widehat{p}_n - p \geq t) \leq e^{-2nt^2}$ and the analogous lower-tail bound [10]. A wrong early output at a fixed inspected prefix requires a deviation of at least $d + t_{n,M}$. Applying a union bound on the underlying full tape, without conditioning on the stopping event, therefore gives

$$\mathbb{P}_{d,s}(Z_M^{\mathrm{ord}} \neq s) \leq e^{-2Md^2} + e^{-2} \sum_{\substack{n<M \\ n \text{ dyadic}}} (n/M)^2 e^{-2nd^2}.$$

For any dyadic sequence of positive u values, the intervals $[u/2, u]$ have disjoint interiors and

$$\int_{u/2}^{u} y e^{-2y}\, dy \geq \frac{3}{8} u^2 e^{-2u}.$$

Consequently $\sum_u u^2 e^{-2u} \leq 2/3$. Writing $x = Md^2 \geq 1$ and using $x^2 e^{-2x} \leq e^{-2}$, the preceding error bound is at most $(5/3)e^{-2}x^{-2} < x^{-2}$. For $x < 1$, the probability bound by one suffices.

Since the stopping time takes dyadic values,

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} = 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n\, \mathbb{P}_{d,s}(T_M^{\mathrm{ord}} > n).$$

If $d > 2t_{n,M}$, survival implies a deviation from the true mean of at least $d/2$. Thus

$$\mathbb{E}_{d,s} T_M^{\mathrm{ord}} \leq 1 + \sum_{\substack{n<M \\ n \text{ dyadic}}} n \left[\mathbf{1}_{\{d \leq 2t_{n,M}\}} + 2e^{-nd^2/2}\right].$$

For $x = Md^2 \geq 1$, the first indicator implies $nd^2 \leq 4(1 + \log x)$. Indeed, this is immediate if $nd^2 < 1$; otherwise $\log(M/n) = \log(x/(nd^2)) \leq \log x$. The corresponding dyadic sum is at most $8d^{-2}(1 + \log x)$. Also

$$\int_{u/2}^{u} e^{-y/2}\, dy \geq \frac{1}{2} u e^{-u/2} \quad \Longrightarrow \quad \sum_u u e^{-u/2} \leq 4.$$

The exponential part of the work bound is therefore at most $8d^{-2}$. Since $1 \le d^{-2}$, total expected work is at most $17d^{-2}(1+\log x)$. Finally $T_M^{\mathrm{ord}} \le M$ deterministically; when $x < 1$, the displayed minimum equals M. □

For any finite measure ξ on $(0, 1/2]$, write

$$\eta_\xi(M) = \int \min\{1, (Md^2)^{-2}\}\, \xi(dd), \qquad O_\xi(M) = \int o_M(d)\, \xi(dd).$$

Both are homogeneous in ξ. The first is nonincreasing in M and the second is nondecreasing. Pointwise comparison also gives $\eta_\xi(4M) \ge \eta_\xi(M)/16$.

Lemma A.2 (A homogeneous dyadic kernel estimate)**.** *For every finite measure ξ and integer $K \ge 1$,*

$$\sum_{k=1}^{K} \sqrt{\eta_\xi(4^k) O_\xi(4^k)} \le 208 \int \min\{2^K, d^{-1}\}\, \xi(dd).$$

In particular,

$$\sum_{k\ge 1} \sqrt{\eta_\xi(4^k) O_\xi(4^k)} \le 208 \int d^{-1}\, \xi(dd), \tag{44}$$

with extended-value interpretation.

Proof. Use bands $(2^{-j-1}, 2^{-j}]$, $j \ge 1$, with masses w_j, and put $c_j = 2^j w_j$. Direct substitution into the two kernels yields

$$\eta_\xi(4^k) \le 2^{-k}\left[\sum_{j\ge k} c_j 2^{k-j} + 16 \sum_{j<k} c_j 2^{3(j-k)}\right] = 2^{-k} A_k,$$

$$O_\xi(4^k) \le 2^k \left[\sum_{j\ge k} c_j 2^{k-j} + 136 \sum_{j<k} c_j 2^{j-k}(1+k-j)\right] = 2^k D_k.$$

For the second display on $j < k$, use $d^{-2} \le 4\, 2^{2j}$ and $1 + \log(4^k d^2) \le 2(1+k-j)$. On $j \ge k$, the deterministic cap gives the first sum exactly as an upper bound.

Apply $\sqrt{A_k D_k} \le (A_k + D_k)/2$ and sum over $k \le K$. For a fixed $j \le K$, the three relevant geometric sums obey

$$\sum_{k\le j} 2^{k-j} \le 2, \qquad \sum_{r\ge 1} 2^{-3r} = \frac{1}{7}, \qquad \sum_{r\ge 1} 2^{-r}(1+r) = 3.$$

The coefficient of c_j is thus at most $2 + 8/7 + 204 < 208$. For $j > K$, only the first kernels occur, with coefficient at most $2\, 2^{K-j}$. Hence the result is at most

$$208 \sum_{j\le K} 2^j w_j + 2\, 2^K \sum_{j>K} w_j.$$

For $j \le K$, $\min\{2^K, d^{-1}\} \ge 2^j$ on the band; for $j > K$ it equals 2^K. This proves the finite bound. Increasing K and using monotone convergence gives (44). All rearrangements involve nonnegative terms, so normalization and moment assumptions are unnecessary. □

A.2 A weighted bank compared with the full null mixture

For a binary prefix $x_{1:n}$, let $P_{p,n}(x_{1:n})$ be its probability under iid Ber(p) observations, and let $P_{\mathrm{f},n} = P_{1/2,n}$. Choose any nonempty finite or countable set of canonical bands

$$I_j = (b_j/2, b_j], \qquad b_j = 2^{-j-1},$$

and positive weights π_j summing to one on these bands. Set $r_j^- = b_j/2$ and $r_j^+ = b_j/4$, and define

$$L_{j,n}^- = \frac{P_{1/2-r_j^-,n}}{P_{\mathrm{f},n}}, \qquad M_n = \frac{\sum_j \pi_j P_{1/2+r_j^+,n}}{P_{-,\mu,n}}, \qquad P_{-,\mu,n} = \int P_{1/2-d,n}\,\mu(dd).$$

For $0 < \epsilon < 1$, stop negative when any $L_{j,n}^- \geq 1/(\epsilon\pi_j)$, and stop positive when $M_n \geq 1/(\rho\epsilon)$. Give negative crossings priority. A zero denominator triggers a positive stop. Denote the bank stopping time by T_{bank}; it may be infinite outside its selected bands.

Lemma A.3 (Full-mixture likelihood bank)**.** *The probability of a wrong bank stop is at most $2\rho\epsilon$ uniformly over $q \in \mathcal{Q}_\rho$, and its false-positive probability is at most $\rho\epsilon$. These statements concern the event that the bank stops and do not assume termination at every magnitude.*

For $d \in I_j$ in a selected band,

$$\mathbb{E}_{d,-} T_{\mathrm{bank}} \leq \frac{3}{4d^2}[1 + \log(1/\epsilon) + \log(1/\pi_j)]. \tag{45}$$

Suppose a finite number $V(d) > 0$ satisfies

$$\beta_\mu(n) := \int e^{-2nx^2}\,\mu(dx) \leq V(d) \qquad \text{for every integer } n \geq d^{-2}.$$

Then

$$\mathbb{E}_{d,+} T_{\mathrm{bank}} \leq \frac{6}{d^2}\left[1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+ \frac{V(d)}{\rho}\right]. \tag{46}$$

In particular the bank has finite conditional work on every selected band whenever such an envelope is supplied.

Proof. A nonnegative supermartingale Y_n with $Y_0 = 1$ has $\mathbb{P}(\sup_n Y_n \geq a) \leq 1/a$: stopping at the first crossing or a fixed horizon gives $1 \geq \mathbb{E}Y_{T\wedge m} \geq a\,\mathbb{P}(T \leq m)$, and increasing the horizon proves the claim. This elementary maximal inequality also works when the process is a supermartingale only on supported prefixes.

Under a positive coin of gap d, the next multiplier of $L_{j,n}^-$ has mean $1 - 4r_j^- d \leq 1$. Its negative-crossing probability is at most $\epsilon\pi_j$. A countable union bound gives conditional negative-crossing probability at most ϵ, so its positive-sign contribution under any admissible prior is at most $\rho\epsilon$.

Under the full all-negative mixture, M_n is a nonnegative supermartingale on supported prefixes. If a prefix x has positive null probability, conditional expectation sums only children xu with positive null probability; their numerator probabilities sum to at most the numerator at x. This proves the supermartingale inequality even when μ has mass at $d = 1/2$. A zero-denominator prefix has null probability zero. Thus a positive crossing has null-mixture probability at most $\rho\epsilon$. Its actual negative-sign contribution is no larger, since it multiplies the conditional crossing probabilities by $1 - q(d) \leq 1$. The two error claims follow.

For negative work, fix a selected-band gap $d \in I_j$ and $r = r_j^- \in [d/2, d]$, with $r \leq 1/4$. The matching log likelihood has mean increment

$$g(d, r) = \frac{1}{2}\log(1 - 4r^2) + d\log\frac{1 + 2r}{1 - 2r} \geq -\frac{8}{3}r^2 + 4dr \geq \frac{4}{3}d^2.$$

Here $\log(1-x) \geq -x/(1-x)$ and $\log((1+x)/(1-x)) \geq 2x$ establish the first lower bound; the concave quadratic takes the value $4d^2/3$ at both endpoints $r = d/2, d$. Every upward log increment is at most $\log(3/2) < 1$. For the first correct-reference crossing τ, the stopped-sum identity at $\tau \wedge m$ gives

$$g(d,r)\mathbb{E}(\tau \wedge m) = \mathbb{E}\log L^-_{j,\tau\wedge m} \leq 1 + \log(1/(\epsilon\pi_j)).$$

Monotone convergence proves integrability of τ and the stated work bound, since $T_{\text{bank}} \leq \tau$. No unbounded stopping identity is invoked before integrability is established.

For positive work, let $E_n = \{\widehat{p}_n - 1/2 \geq d/2\}$. On this event, for $x < 1/2$,

$$\log \frac{P_{1/2-x,n}}{P_{\text{f},n}} = \frac{n}{2}\log(1-4x^2) - n(\widehat{p}_n - \tfrac{1}{2})\log\frac{1+2x}{1-2x} \leq -2nx^2.$$

For $x = 1/2$ the likelihood is zero, since E_n entails at least one head. Therefore $P_{-,\mu,n}/P_{\text{f},n} \leq \beta_\mu(n)$. The matching positive reference $r = r_j^+ \in [d/4, d/2]$ satisfies, on the same event,

$$\log \frac{P_{1/2+r,n}}{P_{\text{f},n}} \geq n\left[-\frac{8}{3}r^2 + 2dr\right] \geq nd^2/3.$$

Set

$$U = 1 + \log(1/\epsilon) + \log(1/\pi_j) + \log_+(V(d)/\rho), \qquad n_0 = \left\lceil 3d^{-2}U \right\rceil.$$

For every $n \geq n_0$, E_n implies $M_n \geq \pi_j e^{nd^2/3}/V(d) \geq 1/(\rho\epsilon)$. Thus $T_{\text{bank}} > n$ implies E_n^c, whose positive-coin probability is at most $e^{-nd^2/2}$. Summing the survival probabilities yields

$$\mathbb{E}_{d,+}T_{\text{bank}} \leq 3d^{-2}U + 2 + 2d^{-2} \leq 6d^{-2}U.$$

The last inequality uses $U \geq 1$ and $d^2 \leq 1/4$.

For a countable bank, the positive mixture is a consistent probability law and the negative stopping event is a measurable countable union. At any fixed time only finitely many negative references can cross: $L^-_{j,n} \leq (3/2)^n$ requires $\pi_j \geq 1/[\epsilon(3/2)^n]$ for a crossing. All preceding arguments therefore apply to finite and countable banks alike. □

A.3 Racing procedures and retaining a genuine unit

Lemma A.4 (Observable race)**.** *Suppose two classifiers run on the odd and even subsequences of one unit's observation tape, with independent auxiliary seeds. Stop the physical procedure when either classifier stops and use its output. On completed counterfactual subsequences its work satisfies*

$$T_{\text{race}} \leq 2\min\{T_1, T_2\}.$$

Its error event is contained in the union of the component wrong-stop events. If either component has a finite deterministic cap, the race terminates with a finite deterministic cap.

Proof. By physical time $2n$, each component has received n observations. If either component would stop after n of its own observations, the race stops no later than $2n$. This proves the pathwise work bound, including an infinite stopping time for the other component. The selected output is wrong only if the component that produces it makes a wrong stop. Conditional on the latent (D, S), both subsequences are iid Bernoulli tapes of the correct mean, so all component error and work bounds apply. The policy uses only the physical observed prefix and its seeds. □

A.4 Finite multilevel allocation at every positive budget

Lemma A.5 (Independent Poisson allocation)**.** *Let $Z_0, \dots, Z_J \in \{0,1\}$ be observable single-unit classifiers, each with finite deterministic cap. Suppose uniformly over $q \in \mathcal{Q}_\rho$ that their error probabilities are at most e_ℓ, their false-positive probabilities are at most f_ℓ, and their expected work is at most $C_\ell > 0$. Put*

$$v_0 = \rho + f_0, \qquad v_\ell = e_\ell + e_{\ell-1} \quad (\ell \geq 1), \qquad k_0 = C_0, \qquad k_\ell = C_\ell + C_{\ell-1} \quad (\ell \geq 1).$$

Assume $v_\ell > 0$ and set $A_J = \sum_{\ell=0}^{J} \sqrt{v_\ell k_\ell}$. For every $B > 0$ there is an observable population estimator of uniform expected work at most B and worst RMS error at most

$$\sqrt{e_J^2 + A_J^2/B}. \tag{47}$$

Proof. A level-zero correction uses a fresh genuine unit and returns $\Delta_0 = Z_0$. For $\ell \geq 1$, a correction draws one fresh latent unit, runs Z_ℓ on its observations, then runs $Z_{\ell-1}$ with a fresh seed on the next unused observations of the same unit, and returns $\Delta_\ell = Z_\ell - Z_{\ell-1}$.

Conditional on the latent state, the unused suffix after a finite stopping time is an iid tape with the same Bernoulli mean: for each possible stopped prefix, independence of its subsequent coordinates gives the claim, and summing over the stopped prefixes preserves it. This conditional argument is applied to successive unused blocks of the same latent unit, even though the first block has random length. The second classifier therefore has its original marginal distribution under the same latent state. No new magnitude or sign is drawn between the two classifiers. Their disagreement implies that at least one misclassifies that common sign, and hence

$$\mathbb{E}_q \Delta_\ell^2 \leq v_\ell, \qquad \mathbb{E}_q W_\ell \leq k_\ell, \qquad \sum_{\ell=0}^{J} \mathbb{E}_q \Delta_\ell = \mathbb{E}_q Z_J.$$

At level zero use $\mathbb{P}_q(Z_0 = 1) \leq \mathbb{P}_q(S = 1) + f_0 \leq \rho + f_0$.

Independently across levels, draw Poisson replication counts with means

$$n_\ell = \frac{B}{A_J} \sqrt{\frac{v_\ell}{k_\ell}}.$$

For each replication use an independent genuine unit, and return

$$\widehat{\theta} = \sum_{\ell=0}^{J} \frac{1}{n_\ell} \sum_{i=1}^{N_\ell} \Delta_{\ell,i}, \qquad N_\ell \sim \text{Poisson}(n_\ell).$$

Its mean is $\mathbb{E}_q Z_J$, so its absolute bias is at most e_J. The expected work is at most $\sum_\ell n_\ell k_\ell = B$.

For independent marks Δ and a Poisson count of mean n, conditioning on the count gives $\text{Var}(\sum_{i=1}^{N} \Delta_i) = n\mathbb{E}\Delta^2$. Consequently the independent level variances sum to at most

$$\sum_{\ell=0}^{J} \frac{v_\ell}{n_\ell} = \frac{A_J^2}{B}.$$

Bias squared plus variance proves (47). All counts are almost surely finite, all component caps are finite, and all first and subsequent observations are included in W_ℓ. □

The allocation may use Poisson means below one; this is valid for every $B > 0$. The bound is uniform over the original rarity class. Finite-level bounds can be optimized by taking an infimum over finite constructions without asserting the existence of a single infinite-level algorithm.

B One-unit null-work obstructions

This appendix supplies the sign-testing and annular ingredients used in Section 4. The weighted sign-testing mechanism is an adaptation of Chen–Li–Qiao's gap-entropy argument, including their Bernoulli extension [5, Lemma 4.1 and Remark D.4]. We give the full finite-gap version needed here, followed by the scale-sensitive gate and its actual minimum-band-mass constraint.

B.1 Weighted sign testing

Lemma B.1 (Weighted sign entropy)**.** *Let $d_1 > \cdots > d_K > 0$, with $d_j/d_{j+1} \geq 2$ and $d_1 \leq 1/128$. One observable binary sign test, with the same independent seed law at all gaps, has error at most $1/10$ under every Bernoulli law $1/2 \pm d_j$. Suppose its stopping time τ has finite expectation under both signs at every such gap. Put $c_j = d_j^2 \mathbb{E}_{-d_j}\tau$. For every probability vector π on these gaps,*

$$\sum_{j=1}^{K} \pi_j c_j \geq \frac{1 + \mathrm{Ent}(\pi)}{214}. \tag{48}$$

No termination or correctness assumption under the fair coin is required.

Proof. First, the ordinary stopped change-of-measure inequality gives a uniform positive floor for each c_j. The terminal sign decision has minus/plus KL divergence at least

$$\mathrm{kl}(9/10, 1/10) = \frac{4}{5}\log 9 > \frac{8}{5}.$$

One Bernoulli observation has divergence at most

$$\frac{(2d_j)^2}{(1/2 + d_j)(1/2 - d_j)} \leq \frac{64}{3} d_j^2.$$

The stopped chain rule, justified by finite expected endpoint work, implies $c_j > 3/40 > 1/16$.

Set $b_0 = 1/1024$ and

$$n_j = \left\lfloor b_0/d_j^2 \right\rfloor, \qquad u_j = \left\lceil 4\mathbb{E}_{-d_j}\tau \right\rceil.$$

Since $d_j \leq 1/128$, we have $n_j \geq b_0/(2d_j^2) \geq 8$. The KL divergence between the complete prefixes of length n_j at the minus and plus laws is at most $1/48$. These prefixes include the independent seed and unused bits if the test has already stopped. Pinsker's inequality therefore gives total variation less than $1/8$. An early minus is a plus-law error, so its probability under the minus law is at most $1/10 + 1/8$. Markov's inequality gives $\mathbb{P}_{-d_j}(\tau > u_j) \leq 1/4$. Consequently the event

$$E_j = \{\text{output minus},\ n_j < \tau \leq u_j\}$$

has $\mathbb{P}_{-d_j}(E_j) > 1/4$.

Let $\mathbb{P}_0$ denote the fair-coin law with the same seed and policy. Comparing deterministic prefixes of length u_j gives KL at most $4d_j^2 u_j \leq 16c_j + 1$. Binary data processing and $\mathrm{kl}(p, z) \geq p\log(1/z) - \log 2$ yield

$$\mathbb{P}_0(E_j) \geq \exp(-64c_j - 8). \tag{49}$$

Only deterministic finite prefixes have been compared with the fair law.

We next bound a weighted overlap pathwise. On a path with finite terminal time t, participation in E_j implies

$$t > n_j \geq \frac{b_0}{2d_j^2}, \qquad t \leq 4\mathbb{E}_{-d_j}\tau + 1.$$

In particular $t \geq 2$ and $c_j \geq (t-1)d_j^2/4 \geq td_j^2/8$. Order the participating gaps from smallest to largest. Their squared values grow by at least a factor of four, and the smallest has $td_j^2 > b_0/2$. The corresponding costs are thus bounded below by $b_0 4^r/16 = 4^{r-7}$, for $r = 0, 1, \ldots$. Hence

$$\sum_j \frac{\mathbf{1}_{E_j}}{1+c_j} \leq \sum_{r\geq 0} \frac{1}{1+4^{r-7}} \leq 7 + \sum_{r\geq 7} 4^{7-r} = \frac{25}{3}.$$

On a nonterminating fair path no E_j occurs, so the same inequality holds. Integrating and using (49), together with $1 + c_j \leq e^{c_j}$, gives

$$Z := \sum_j e^{-65c_j} \leq (25/3)e^8, \qquad \log Z < 11.$$

Gibbs' inequality for π and the normalized masses e^{-65c_j}/Z implies

$$\mathrm{Ent}(\pi) \leq 65 \sum_j \pi_j c_j + \log Z.$$

Writing $\bar{c} = \sum_j \pi_j c_j$ and retaining the stronger floor $\bar{c} \geq 3/40$, we obtain

$$1 + \mathrm{Ent}(\pi) \leq 65\bar{c} + 9 + \log(25/3) \leq \left[65 + \frac{40}{3}\{9 + \log(25/3)\}\right] \bar{c} < 214\bar{c}.$$

The windows need not be disjoint; only the displayed weighted-overlap bound is used. □

B.2 A moment comparison on the half-prior class

Write $X_\rho = \{0 \leq q \leq 1/2 : \int q\, d\mu \leq \rho\}$. For an integrable observable one-unit output Y with finite uniform variance and uniform bias at most $\rho/64$, put $V = \sup_{q\in X_\rho} \mathrm{Var}_q\, Y$ and $M = \sup_{q\in X_\rho} \mathbb{E}_q Y^2$.

Lemma B.2 (Moment comparison)**.** *For $0 < \rho \leq 1/4$, one has $V \geq 2\rho/3$ and $M \leq 3V/2$.*

Proof. Let $f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y$. Comparing the null population with the constant prior $q = \rho$ gives $\int f\, d\mu \geq 31/32$. Conditional variance at this constant prior and Jensen's inequality yield

$$V \geq \rho(1-\rho) \int f^2\, d\mu \geq \rho(1-\rho)(31/32)^2 \geq 2\rho/3.$$

Every mean has absolute value at most $65\rho/64$, so $M \leq V + (65\rho/64)^2 \leq 3V/2$. The constant prior also gives integrability of both fixed-sign second moments; all conditional statements may be taken outside a common null set. □

B.3 An activation gate that respects the magnitude scale

The new lower-bound ingredient charges activation at the information scale $a^2\mathbb{E}_0 T$. In this subsection μ is any probability law supported on $(0, a]$, $a \leq 1/4$, and

$$\mathcal{X}_\rho = \{q : 0 \leq q \leq \tfrac{1}{2},\ \textstyle\int q\, d\mu \leq \rho\}.$$

Let (Y, T) be an observable one-coin procedure with uniform bias at most $\rho/64$, finite uniform variance V, and finite expected work under every $q \in \mathcal{X}_\rho$. Write

$$M_0 = \sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T, \qquad P = M_0 W_0.$$

Here M_0 is the uniform second-moment bound, whereas W_0 is null work; the subscript on M_0 does not restrict its supremum to the null. The moment comparison in Lemma B.2 gives $M_0 \leq 3V/2$.

Lemma B.3 (A scale-sensitive baseline and gated activation)**.** *Put*

$$C_g = e^7, \qquad \kappa = 64/3, \qquad c_g = (256\kappa C_g^4)^{-1}.$$

Every procedure above satisfies

$$P \geq \frac{\rho}{64a^2}. \tag{50}$$

If $a^2P \leq c_g$, there is an observable procedure Z on the same one-coin experiment, with moment bound m_Z, null work W_0^Z, and a measurable set $\mathcal{G}$ of good magnitudes, such that

$$\begin{gathered}\sup_{q\in\mathcal{X}_\rho} \mathbb{E}_q Z^2 \leq m_Z, \qquad \rho/16 \leq m_Z \leq 2C_g^4 a^2 P, \\ m_Z W_0^Z \leq C_g^4 P, \qquad \mu(\mathcal{G}^c) \leq 144 m_Z.\end{gathered} \tag{51}$$

At every good gap, both signs have finite conditional expected work and

$$|\mathbb{E}[Z \mid d,0]| \leq \frac{1}{8}, \qquad \mathbb{E}[Z \mid d,1] \geq \frac{1}{2}, \qquad \mathbb{E}[Z^2 \mid d,0] \leq \frac{m_Z}{\rho}, \qquad \mathbb{E}[Z^2 \mid d,1] \leq \frac{3m_Z}{\rho}. \tag{52}$$

One common sign test with both errors below $1/10$ on these gaps uses

$$n = \lceil 1024 m_Z/\rho \rceil \leq 1040 m_Z/\rho \tag{53}$$

independent restarts of Z on unused observations of the same coin. Restart independence is conditional on its fixed latent magnitude and sign, not after averaging over the population. Its negative expected work is $n\,\mathbb{E}[T_Z \mid d,0]$.

Proof. We first prove (50). Let P_- and P_+ be the full transcript laws with uniformly negative and uniformly positive signs, respectively. The priors $q = 0$ and $q \equiv \rho$ give

$$\mathbb{E}_+ Y - \mathbb{E}_- Y \geq 31/32, \qquad \mathbb{E}_- Y^2 \leq M_0, \qquad \mathbb{E}_+ Y^2 \leq M_0/\rho.$$

The positive moment and expected work are finite because $q \equiv \rho$ is admissible. Revealing D for the information calculation, and using the stopped chain rule, gives

$$h(P_-, P_+) \leq \tfrac{1}{2}\kappa a^2 W_0,$$

where $h = 1 - \mathrm{Aff}$. Indeed the one-observation endpoint divergence is at most κd^2 for $d \leq 1/4$. For any square-integrable output F,

$$|\mathbb{E}_P F - \mathbb{E}_Q F|^2 \leq 4(\mathbb{E}_P F^2 + \mathbb{E}_Q F^2) h(P,Q). \tag{54}$$

Thus

$$(31/32)^2 \leq (5\kappa/2) a^2 P/\rho,$$

which implies (50).

The decision to take a first observation is seed-only and precedes all data. The information gate below is instead an observable finite-prefix stopping rule; neither step reveals a latent magnitude. For the gate, on a full observation tape define

$$S_n = \sum_{i=1}^{n} (2X_i - 1), \qquad n_0 = \lceil a^{-2} \rceil, \qquad H_0 = \lceil a^{-1} \rceil,$$

$$G = \min\{n_0, \inf\{n \geq 1 : |S_n| \geq H_0\}\}, \qquad \sigma = T \wedge G, \qquad E = \{T \geq G\}. \tag{55}$$

The event E is measurable at σ, including the tie $T = G$. Include the entire independent seed in the stopped prefix. At σ, $|S_\sigma| \le H_0$ and $\sigma \le n_0$. For a fixed $d \le a$ and sign s, the likelihood relative to the fair stopped-prefix law is

$$\exp\left\{(2s-1)S_\sigma \operatorname{arctanh}(2d) + \frac{\sigma}{2}\log(1-4d^2)\right\}.$$

The inequalities

$$\operatorname{arctanh}(2d) \le 8a/3, \qquad -\tfrac{1}{2}\log(1-4d^2) \le 8a^2/3$$

show that the absolute logarithm is at most $37/6 < 7$. Every fixed-state prefix law, and the all-negative magnitude mixture, therefore has density in $[C_g^{-1}, C_g]$ relative to fair.

Let $p = \mathbb{P}_{\mathrm{fair}}(E)$ and $p_s(d) = \mathbb{P}_{d,s}(E)$. Then

$$p/C_g \le p_s(d) \le C_g p. \tag{56}$$

Bounded fair stopping gives $\mathbb{E}_{\mathrm{fair}} S_\sigma^2 = \mathbb{E}_{\mathrm{fair}}\sigma$. A gate activation either reaches n_0 or hits H_0, so

$$p \le \left(n_0^{-1} + H_0^{-2}\right)\mathbb{E}_{\mathrm{fair}}\sigma \le 2a^2\mathbb{E}_{\mathrm{fair}}\sigma \le 2C_g a^2 W_0. \tag{57}$$

Set $Y_e = Y\mathbf{1}_{\{T<G\}}$, explicitly zero on E. This is a function of the bounded prefix. Its second moment under either fixed state is at most $C_g^2 M_0$. The stopped endpoint divergence and the same prefix domination give

$$\mathrm{KL}(P^\sigma_{-d}, P^\sigma_{+d}) \le \kappa d^2 \mathbb{E}_{-d}\sigma \le \kappa C_g^2 a^2 W_0.$$

Since $h \le \mathrm{KL}/2$, (54) implies

$$|\mathbb{E}_{+d}Y_e - \mathbb{E}_{-d}Y_e|^2 \le 4\kappa C_g^4 a^2 P. \tag{58}$$

Under $a^2 P \le c_g$, every such early mean difference is at most $1/8$ in absolute value. In this branch $p > 0$: if $p = 0$, the gate never activates under any state, so $Y = Y_e$, contradicting the integrated mean separation $31/32$ from $q = 0, q \equiv \rho$.

Here is an actual observable procedure realizing the conditioning. Keep the same genuine coin. Repeatedly run the original procedure, using a fresh original seed and unused iid flips for each trial. Reject a trial if its original stopping time is before its gate. When a trial reaches the gate, accept it, continue that original trial to its original stopping time, and return

$$Z = C_g p Y.$$

Every rejected observation is charged. Conditional on d, s, trials are iid with acceptance probability $p_s(d) > 0$. Writing $w_s(d) = \mathbb{E}[T \mid d, s]$, the renewal identity gives

$$\mathbb{E}[T_Z \mid d, s] = \frac{w_s(d)}{p_s(d)}. \tag{59}$$

This identity includes the work of all rejected trials and the final accepted trial. Moreover,

$$\mathbb{E}[Z^2 \mid d, s] = \frac{C_g^2 p^2}{p_s(d)}\mathbb{E}[Y^2\mathbf{1}_E \mid d, s] \le C_g^3 p\,\mathbb{E}[Y^2 \mid d, s].$$

With $m_Z = C_g^3 p M_0$, integration and (57)–(59) give

$$\sup_q \mathbb{E}_q Z^2 \le m_Z, \qquad m_Z \le 2C_g^4 a^2 P, \qquad W_0^Z \le C_g W_0/p, \qquad m_Z W_0^Z \le C_g^4 P. \tag{60}$$

Expected work under every q is finite by the same comparison.

We establish conditional sign separation directly; conditioning has not preserved the estimator's means. Write

$$f_s(d) = \mathbb{E}[Y \mid d, s], \quad \ell_s(d) = \mathbb{E}[Y\mathbf{1}_E \mid d, s], \quad z_s(d) = \mathbb{E}[Z \mid d, s].$$

The set $\{f_+ - f_- < 3/4\}$ has mass below 2ρ. Otherwise a fractional prior of total mass ρ on that set would contradict the bias guarantee and the mean separation $31\rho/32$. Outside that set, (58) gives $\ell_+ - \ell_- \geq 5/8$. Also

$$z_s(d) = \frac{C_g p}{p_s(d)}\ell_s(d), \qquad 1 \leq \frac{C_g p}{p_s(d)} \leq C_g^2.$$

If $|z_-(d)| \leq 1/8$, then $|\ell_-(d)| \leq 1/8$, $\ell_+(d) \geq 1/2$, and $z_+(d) \geq 1/2$. The exceptional mass for $|z_-| > 1/8$ is at most $64m_Z$. Since

$$m_Z \leq 2C_g^4 c_g = 1/(128\kappa) < 1/256,$$

this preliminary mean-good set has mass at least $1 - 2\rho - 64m_Z \geq 1/4$. Taking $q \equiv \rho$ in the second-moment bound yields $m_Z \geq \rho/16$.

Finally, put $a_Z(d) = \mathbb{E}[Z^2 \mid d, 0]$ and $b_Z(d) = \mathbb{E}[Z^2 \mid d, 1]$. The set $\{a_Z > m_Z/\rho\}$ has mass at most ρ. The set $\{b_Z - a_Z > 2m_Z/\rho\}$ has mass below 2ρ, again by using a fractional admissible prior on a set of mass at least 2ρ. Removing these sets and null sets of infinite conditional work gives total exceptional mass

$$5\rho + 64m_Z \leq 144m_Z$$

and proves (52). Average n independent restarts of Z on unused blocks of the same coin, and threshold at $5/16$. Both mean margins are at least $3/16$, so Chebyshev gives errors at most

$$\frac{3m_Z/\rho}{n(3/16)^2} \leq 1/12 < 1/10.$$

The bound on n follows from $m_Z \geq \rho/16$, and work adds, proving (53). □

B.4 A scaled annular obstruction

For a positive vector $v = (v_i)$, set

$$\mathcal{G}(v) = A[1 + \mathrm{Ent}(v/A)], \qquad A = \sum_i v_i.$$

A useful elementary identity is

$$\frac{\partial \mathcal{G}}{\partial v_i} = 1 + \log(A/v_i) \geq 1. \tag{61}$$

Thus this cost-entropy functional is coordinatewise increasing. In particular, if $u \leq v_i \leq U$ for K coordinates, then

$$uK(1 + \log K) \leq \mathcal{G}(v) \leq UK(1 + \log K). \tag{62}$$

Proposition B.4 (A gated annular lower bound)**.** *Suppose μ is supported on $(0, a]$, $a \leq 1/4$. Take a nonempty finite annular family*

$$I_j = (b_j/2, b_j], \qquad b_j/b_{j+1} \geq 4, \qquad b_1 \leq 1/128,$$

with positive masses $w_j = \mu(I_j)$. Define

$$w_* = \min_j w_j, \qquad A = \sum_j w_j/b_j^2, \qquad \pi_j = \frac{w_j/b_j^2}{A}, \qquad \Gamma^\circ = A[1 + \mathrm{Ent}(\pi)].$$

Every one-coin procedure with the assumptions above satisfies

$$V\mathbb{E}_0 T \geq \frac{e^{-28}}{2\,000\,000} \min\{\rho\Gamma^\circ, w_*/a^2\}. \tag{63}$$

Proof. If $a^2P \geq c_g$, use $w_* \leq 1$ and $M_0 \leq 3V/2$. Otherwise apply Lemma B.3. If $m_Z > w_*/288$, then

$$P > w_*/(576C_g^4a^2).$$

If $m_Z \leq w_*/288$, at least half the mass of every annulus is good. Averaging selects $d_j \in I_j \cap \mathcal{G}$ with

$$w_-^Z(d_j) \leq \frac{4}{w_j}\int_{I_j} w_-^Z(d)\,\mu(dd).$$

The selected gaps are separated by a factor greater than two and bounded by $1/128$. Apply Lemma B.1 to the common restart test, with weights π_j:

$$\frac{1+\mathrm{Ent}(\pi)}{214} \leq n\sum_j \pi_j d_j^2 w_-^Z(d_j) \leq \frac{4n}{A}W_0^Z.$$

Consequently,

$$\rho\Gamma^\circ \leq 856\rho n W_0^Z \leq 890\,240\, m_Z W_0^Z \leq 890\,240\, C_g^4 P.$$

Converting $P = M_0W_0$ to VW_0 completes all branches with the displayed conservative constant. No atomlessness is used. □

The entropy branch is available when $\rho a^2\Gamma^\circ \leq w_*$. This condition is invariant under a common rescaling of all magnitudes.

C Convex comparison, common pairs, and additive localization

This appendix proves the specialized transfer inputs used by the scalar lower bound and the stopping-program comparison. The finite-experiment convex-estimation principle has antecedents in Juditsky–Nemirovski and Polyanskiy–Wu [6, 7]; the midpoint adaptive decomposition is due to Lee–Valiant [3]. The arguments needed for this model are included in full.

C.1 Finite categorical convex comparison

Lemma C.1 (Finite categorical convex comparison)**.** *Let X be a nonempty compact convex subset of a finite-dimensional real vector space. Let $\ell : X \to \mathbb{R}$ be affine, and let $p(x)$ be a strictly positive categorical probability vector with affine continuous coordinates. For $n \geq 1$ independent observations from $p(x)$, define*

$$\Omega_n = \max\{\ell(x) - \ell(y) : x, y \in X, \mathrm{Aff}(p(x), p(y))^n \geq 1/2\}.$$

For every $\eta > 0$, there is a bounded additive estimator $F = c + \sum_{i=1}^n \phi_i(Z_i)$ satisfying

$$\sup_{x\in X}\{\mathbb{E}_x(F - \ell(x))^2\}^{1/2} \leq \tfrac{3}{2}\Omega_n + \eta. \tag{64}$$

Proof. We reproduce the short argument needed for the numerical constant. Set $s = \log 2$, let $\mathcal{F}$ be the finite-dimensional space of additive functions, and define, for $\alpha > 0$,

$$\Phi(x, y; \phi, \alpha) = \ell(x) - \ell(y) + \alpha\log\mathbb{E}_y e^{\phi/\alpha} + \alpha\log\mathbb{E}_x e^{-\phi/\alpha} + 2\alpha s.$$

The function is continuous and concave in (x, y): independence writes each log moment as a sum of logarithms of positive affine forms. It is convex in (ϕ, α) by the perspective transformation of a log moment generating function. Sion's minimax theorem [11] therefore applies to $X \times X$ and $\mathcal{F} \times (0, \infty)$. Write $2S = \inf_{\phi,\alpha>0}\max_{x,y}\Phi$.

For a fixed pair, the half log likelihood ratio belongs to $\mathcal{F}$ and Cauchy–Schwarz gives

$$\inf_{\psi\in\mathcal{F}}\left[\log \mathbb{E}_y e^{\psi} + \log \mathbb{E}_x e^{-\psi}\right] = 2\log \mathrm{Aff}(p(x)^{\otimes n}, p(y)^{\otimes n}).$$

Substituting $\phi = \alpha\psi$ and minimizing over $\alpha > 0$ gives the value $\ell(x) - \ell(y)$ when the affinity is at least $1/2$, and $-\infty$ otherwise. Consequently $2S = \Omega_n$. The constrained maximum exists because its feasible set is nonempty and compact.

For $R > S$, choose (ϕ, α) with $\max \Phi \le 2R$. On the diagonal, Cauchy–Schwarz implies $\Phi(x, x; \phi, \alpha) \ge 2\alpha s$, hence $\alpha \le R/s$. Set

$$U = \max_x\{\ell(x) + \alpha \log \mathbb{E}_x e^{-\phi/\alpha}\}, \qquad V = \max_x\{-\ell(x) + \alpha \log \mathbb{E}_x e^{\phi/\alpha}\}.$$

Then $U + V + 2\alpha s \le 2R$. The estimator $F = \phi + (U - V)/2$ satisfies

$$\mathbb{E}_x e^{\pm(F-\ell(x))/\alpha} \le e^{R/\alpha - s}.$$

Therefore $\mathbb{P}_x\{|F - \ell(x)| > R + t\} \le e^{-t/\alpha}$ for $t \ge 0$. Tail integration yields

$$\mathbb{E}_x(F - \ell(x))^2 \le R^2 + 2R\alpha + 2\alpha^2 \le R^2\left(1 + \frac{2}{\log 2} + \frac{2}{(\log 2)^2}\right) < 9R^2.$$

Let $R \downarrow S$, with the slack chosen for η. Every component is bounded because its alphabet is finite. This proves (64). □

C.2 Midpoint decomposition for interleaved observations

We use $h(P, Q) = 1 - \mathrm{Aff}(P, Q)$, so $0 \le h \le 1$ and $\mathrm{TV}(P, Q) \le \sqrt{2h(P, Q)}$. The following finite-prefix argument is the midpoint decomposition of Lee–Valiant [3, Lemmas 17–18], written in the form needed here.

Lemma C.2 (Midpoint work decomposition)**.** *Fix two endpoint priors q_-, q_+ and their midpoint $\bar{q} = (q_- + q_+)/2$. Suppose every capped randomized one-unit policy A obeys*

$$h(P_-^A, P_+^A) \le \kappa\, \mathbb{E}_{\bar{q}} T_A.$$

Then every population algorithm with a deterministic total cap satisfies

$$h(P_-^{\mathrm{pop}}, P_+^{\mathrm{pop}}) \le \kappa\, \mathbb{E}_{\bar{q}} T_{\mathrm{pop}}.$$

It suffices that the local hypothesis hold up to that same total cap. If instead $q_\pm \le 1/2$ and the local bound uses null work $\lambda\mathbb{E}_0 T_A$, the conclusion holds with $\kappa = 2\lambda$. The statement also holds for finitely many observable component types, each with its own endpoint pair and midpoint, when the same coefficient bounds every local policy for every type. Types may be selected adaptively, but each fresh unit's type is fixed before its first observation and retained on every revisit.

Proof. Fix total cap K and condition on the algorithm's independent seed. Defer latent draws until a unit is first observed, which changes no observable law. Index units by that first observation. A consistent global leaf z contains local binary words $h_i(z)$, with empty words for unused slots. The endpoint leaf densities factor as

$$p_\pm^{\mathrm{pop}}(z) = \prod_{i=1}^{K} p_\pm(h_i(z)).$$

The common policy-consistency indicator is implicit. Put $m_i = (p_-(h_i) + p_+(h_i))/2 = p_{\bar{q}}(h_i)$ and $c_i = \sqrt{p_-(h_i)p_+(h_i)}$. Then

$$\prod_i m_i - \prod_i c_i \le \sum_i (m_i - c_i) \prod_{j \ne i} m_j.$$

The integral of $\prod_i m_i$ over consistent leaves is one, since it is the population law in which every fresh unit has midpoint prior. Integrating the inequality therefore bounds population squared Hellinger by a sum of hybrid squared Hellinger distances. In hybrid i only unit i has an endpoint prior; all other units have midpoint priors.

Simulate all other units from independent midpoint tapes included in an auxiliary seed, and run the population rule with one genuine unit in slot i. This is a capped local policy. Its full transcript, including that seed, determines the hybrid global transcript. Data processing bounds hybrid distance by local distance. The expected midpoint local works sum to $\mathbb{E}_{\bar{q}} T_{\mathrm{pop}}$ because each purchased observation belongs to exactly one slot. This proves the first assertion, including after integrating the original seed. The charged expectation is under the midpoint population, admissible by convexity. It is not replaced by the average endpoint work, which can be much smaller under an adaptive stopping rule.

If $q_\pm \le 1/2$, the midpoint latent law dominates half the null latent law. The same measure inequality holds after passing through any observation policy and auxiliary seed. Thus $\mathbb{E}_0 T_A \le 2\mathbb{E}_{\bar{q}} T_A$.

For observable types, use the type selected at a slot in each leaf. The same product factorization holds, with its type-specific local kernel. To realize a hybrid locally, first simulate the midpoint history before the slot's first observation; its type and that history are part of the independent seed. The genuine unit has the selected type's prescribed latent law independently of that seed. Future simulated midpoint units use fresh independent seeds. Conditional on the preceding history, the hypothesis for the selected type applies. Sum and integrate as above. No latent magnitude or sign is supplied to the local policy. □

For an unbounded stopping rule, apply the conclusion to every bounded global prefix. Total variation of the full transcript is the increasing limit of the prefix total variations: bounded cylinder functions generate the transcript sigma-field, or equivalently one applies conditional expectation to the likelihood densities under the dominating sum law. Finite expected endpoint work ensures almost-sure finite termination. A common pair valid for every cap therefore gives the same terminal testing bound. Equivalently, couple the stopped experiment with its cap-L truncation under each endpoint. Their total coupling error is at most $\mathbb{P}_-(T > L) + \mathbb{P}_+(T > L)$, which tends to zero. This passage requires no common deterministic bound on stopping.

C.3 The half-prior class and null work

Put

$$X_\rho = \left\{ q : 0 \le q \le \tfrac{1}{2}, \quad \theta(q) := \int q \, d\mu \le \rho \right\}, \qquad 0 < \rho \le \tfrac{1}{4}.$$

For a capped observable one-coin procedure with bounded real output Y, write

$$V = \sup_{q \in X_\rho} \mathrm{Var}_q \, Y, \qquad M = \sup_{q \in X_\rho} \mathbb{E}_q Y^2, \qquad W_0 = \mathbb{E}_0 T,$$

and define

$$\mathfrak{J}_0(b) = \inf \left\{ VW_0 : \sup_{q \in X_\rho} |\mathbb{E}_q Y - \theta(q)| \le b, \ T \text{ capped}, \ Y \text{ bounded} \right\}. \tag{65}$$

An empty infimum is infinity. The function is nonincreasing in b, and $\mathfrak{J}_0(\rho/2) = 0$.

C.4 Transfer at an arbitrary bias tolerance

Theorem C.3 (Bias-indexed common pairs and population lower bounds)**.** *Suppose that every capped bounded one-coin output with uniform bias at most $b > 0$ over X_ρ satisfies $VW_0 \geq J$, where $0 < J < \infty$. Then, for every $0 < \delta \leq b/3$, there is one pair $q_-, q_+ \in X_\rho$ with target separation at least δ such that simultaneously for all capped randomized local policies A,*

$$h(P^A_{q_-}, P^A_{q_+}) \leq 42\frac{\delta^2}{J}\mathbb{E}_0 T_A. \tag{66}$$

Consequently,

$$R_{\mu,\rho}(B) \geq \frac{1}{128}\min\{b, \sqrt{J/B}\}. \tag{67}$$

The conclusion includes fully adaptive creation, revisiting, interleaving, nonlinear outputs and unbounded stopping.

Proof. The premise forces $b < \rho/2$, since the zero-work constant $\rho/2$ would otherwise contradict it. First fix a finite strictly positive categorical statistic of a capped policy of null work $w > 0$. For n independent runs, let $F = c + \sum_{i=1}^n Y_i$ be a bounded additive estimator of worst RMS r. Put $b_i = \mathbb{E}_0 Y_i$, $b_F = \mathbb{E}_0 F$, and choose an independent uniform index I. Running just its component and returning $Z = n(Y_I - b_I) + b_F$ preserves all population means at null work w.

The laws under q and zero share a subprobability of mass $1 - \theta(q)$, obtained by weighting the negative-sign kernel by $1 - q(d)$. The elementary common-mixture variance inequality gives

$$(\mathbb{E}_q Y_i - b_i)^2 \leq \frac{2\theta(q)}{1-\theta(q)}(\operatorname{Var}_q Y_i + \operatorname{Var}_0 Y_i).$$

Indeed, either mean's displacement from the normalized common part is at most the square root of $\theta/(1-\theta)$ times its standard deviation; square the sum of the two displacements. Independence in F now yields

$$\sup_q \operatorname{Var}_q Z \leq nr^2\left(1 + \frac{4\rho}{1-\rho}\right) \leq \tfrac{7}{3}nr^2.$$

If $r < b$, apply the assumed obstruction to Z; if $r \geq b$, retain that inequality. In both cases,

$$r \geq \min\{b, \sqrt{3J/(7nw)}\}.$$

Lemma C.1 gives an additive estimator of RMS at most $(3/2)\Omega_n + \eta$, where Ω_n is the maximum target separation whose product affinity is at least $1/2$. Letting $\eta \downarrow 0$ therefore gives

$$\Omega_n \geq \tfrac{2}{3}\min\{b, \sqrt{3J/(7nw)}\}.$$

Set $x = J/(21\delta^2 w)$. If $x \geq 2$, take $n = \lfloor x \rfloor$; then $\Omega_n \geq 2\delta$, and the corresponding pair satisfies

$$h \leq \frac{\log 2}{n} \leq 42\log 2\,\frac{\delta^2 w}{J} \leq 42\frac{\delta^2 w}{J}.$$

If $x < 2$, the last right side exceeds one, so any separated pair suffices. If $w = 0$, the seed-only first-observation event has probability zero; all local laws agree and $q_- = 0, q_+ = \delta$ suffices.

We now obtain one pair simultaneously for every capped local policy. For an arbitrary capped policy, partition its full transcript into finitely many measurable cells, then replace the cell independently by a uniform cell with probability $\varepsilon > 0$. Run the policy even on replacement outcomes, so its null work is unchanged. The resulting strictly positive category probabilities and the target have compact convex finite-dimensional range, so the preceding comparison applies. For any finite collection of these experiments, independently mix them and reveal the chosen

index. Both work and squared Hellinger distance are affine in the mixture weights. Hellinger is continuous and convex in the parameter pair. Finite-simplex minimax gives a pair satisfying every constraint in the collection. The constraint sets are closed in the weak-star compact convex set

$$\{(q_-, q_+) \in X_\rho^2 : \theta(q_+) - \theta(q_-) \geq \delta\}.$$

The finite intersection property supplies one pair for all experiments. For fixed b and δ, this pair is chosen before the adaptive population algorithm and its total-observation cap. The same pair works for every capped local policy and every cap. Let $\varepsilon \downarrow 0$, then use that full squared Hellinger distance is the supremum over finite measurable partitions. This proves (66) for arbitrary transcript spaces.

The midpoint decomposition in Lemma C.2 bounds population squared Hellinger by $84\delta^2 B/J$: the local null cost is at most twice the midpoint cost, and summing midpoint costs counts each observation once. Set

$$\delta = \frac{1}{32}\min\{b, \sqrt{J/B}\}.$$

The distance is at most $84/1024 < 1/8$, hence TV is at most $1/2$. The same pair works for every deterministic observation cap. Finite expected endpoint work and increasing capped prefixes give the same terminal TV bound for unbounded stopping. Thresholding an estimator at the target midpoint gives RMS at least $\delta/4$, proving (67). A budget over the full rarity class implies the required budget over X_ρ. □

C.5 A direct variational null-work obstruction

For Bernoulli observations, define the known variational modulus [2]

$$\Psi_{\mu,\rho}(h) = \sup_{\substack{0\leq w\leq 1 \\ \int w\, d\mu \leq \rho}} \int \min\{w(d), h\sqrt{w(d)}/d\}\, \mu(dd). \tag{68}$$

The proof below establishes the exact null-work input required here directly from fixed-sign comparisons. No optimizer formula is needed.

Proposition C.4 (Variational accuracy-dependent null obstruction)**.** *For every probability measure μ on $(0, 1/2]$ and every $0 \leq b \leq \rho/64$,*

$$\mathfrak{J}_0(b) \geq \sup_{h>0} \frac{(\Psi_{\mu,\rho}(h) - 4b)_+^2}{192h^2}. \tag{69}$$

No nonatomicity, finite support, or separation of magnitudes is assumed.

Proof. Fix an eligible (Y, T), and define its conditional response, moments, and negative work by

$$f(d) = \mathbb{E}_{d,+}Y - \mathbb{E}_{d,-}Y, \qquad A_s(d) = \mathbb{E}_{d,s}Y^2, \qquad w_-(d) = \mathbb{E}_{d,-}T.$$

We first show

$$f(d)^2 \leq 64d^2 w_-(d)[A_-(d) + A_+(d)]. \tag{70}$$

For $d \leq 1/4$, the stopped KL chain rule gives $\mathrm{KL}(P^A_{d,-}, P^A_{d,+}) \leq (64/3)d^2 w_-(d)$. For any square-integrable output under two laws,

$$|\mathbb{E}_P Y - \mathbb{E}_Q Y|^2 \leq 4(\mathbb{E}_P Y^2 + \mathbb{E}_Q Y^2)h(P, Q), \qquad h(P, Q) \leq \tfrac{1}{2}\,\mathrm{KL}(P, Q).$$

These prove (70) with coefficient $128/3 < 64$. For $d > 1/4$, the conditional laws agree whenever the seed requests no observation. Their Hellinger distance is therefore at most the activation

probability, hence at most $w_-(d)$. The same mean inequality gives $f(d)^2 \le 4w_-(d)(A_- + A_+) \le 64d^2 w_-(d)(A_- + A_+)$. This also handles $d = 1/2$ without a singular endpoint KL comparison.

For every feasible w in (68), the allocation $q = w/2$ belongs to X_ρ, and

$$\int w(A_- + A_+)\, d\mu \le 2 \int [(1 - w/2)A_- + (w/2)A_+]\, d\mu \le 2M.$$

Weighted Cauchy–Schwarz and (70) yield

$$\left(\int \frac{\sqrt{w}\,|f|}{d}\, d\mu \right)^2 \le W_0 \int \frac{w f^2}{d^2 w_-}\, d\mu \le 128 M W_0. \tag{71}$$

On $\{w_- = 0\}$, the response inequality forces $f = 0$, and the quotient is zero. Truncation justifies Cauchy–Schwarz before passing to the limit if an integral is initially extended-valued.

Put $v_h = \min\{w, h\sqrt{w}/d\}$. Comparing the mean under the admissible allocation $q = v_h/2$ with the null mean, and using both bias bounds, gives

$$\int v_h f\, d\mu \ge \int v_h\, d\mu - 4b.$$

Since $v_h \le h\sqrt{w}/d$, (71) implies

$$M W_0 \ge \frac{(\int v_h\, d\mu - 4b)_+^2}{128 h^2}.$$

Take the supremum over w. To convert the second moment to variance, the null and constant-prior populations force the all-positive and all-negative means to differ by at least $31/32$. The mixture with constant sign prior ρ consequently has variance at least $\rho(1-\rho)(31/32)^2 \ge 2\rho/3$. Also every admissible mean has absolute value at most $65\rho/64$. Thus $M \le V + (65\rho/64)^2 \le 3V/2$, proving (69). □

This product bound accounts for accuracy-dependent tail difficulty in the same framework as the annular scale-entropy obstruction. At a fixed gap d, the directly evaluated identity $\Psi(h) = \min\{\rho, h\sqrt{\rho}/d\}$ recovers product scale ρ/d^2 as $b \downarrow 0$. For the quadratic law it supplies the inverse-square logarithm; the additional scale-entropy factor still requires the annular argument.

Proposition C.5 (An optimizer for the variational modulus)**.** *For $0 < \rho < 1$, there is a unique $\alpha > 0$ satisfying*

$$\int \min\{1, \alpha^2/d^2\}\, \mu(dd) = \rho.$$

The allocation $w_*(d) = \min\{1, \alpha^2/d^2\}$ maximizes $\Psi_{\mu,\rho}(h)$ simultaneously for every $h > 0$. In particular, $\Psi(h) = \rho$ when $h \ge \alpha$. Fractional allocation at an atom is permitted and may be essential; this is distinct from the hard assignment statement for the entropy-selection infimum.

Proof. The mass on the left is continuous in α, tends to zero at zero and to one at infinity, and is strictly increasing while it is below one. These assertions follow by dominated convergence and the positive mass of unsaturated magnitudes. Thus the stated root exists uniquely. If $h \ge \alpha$, the allocation attains the upper bound ρ. If $h < \alpha$, put $\lambda = h/(2\alpha) < 1/2$. The function

$$\phi_d(w) = \min\{w, (h/d)\sqrt{w}\}, \qquad 0 \le w \le 1,$$

is concave. For $d > \alpha$, the point $w_* = (\alpha/d)^2$ is in its square-root branch and has derivative λ. For $h < d \le \alpha$, the left derivative at $w_* = 1$ is $h/(2d) \ge \lambda$; for $d \le h$, $\phi_d(w) = w$. Hence w_* maximizes $\phi_d(w) - \lambda w$ pointwise. Integrating and using its mass ρ proves optimality over all allocations of mass at most ρ, including atomic laws. □

Remark C.6 (The bias subtraction cannot be uniformly reduced). Let $\rho = 1/8$, $\varepsilon = 2^{-10}$, and $\mu = (1-\varepsilon)\delta_{1/2} + \varepsilon\delta_d$. The one-flip output $Y = X - b$, with $b = \varepsilon(1-2d)/4$, has bias at most b on X_ρ, null work one, and variance at most $1/4$. Indeed its bias is $\varepsilon(1/2-d)(1-2q(d)) - b \in [-b, b]$. Yet $\Psi(d) \geq \varepsilon$. Replacing $4b$ by cb with any fixed $c < 4$ in (69) would make its right-hand side diverge as $d \downarrow 0$. The coefficient four cannot be reduced in this form of the bound, even if its denominator constant is changed.

C.6 A direct sum of conditional experiments

Let $\sigma_1, \ldots, \sigma_m$ be nonzero finite submeasures with $\sum_i \sigma_i \leq \mu$. Write $\tau_i = \sigma_i((0, 1/2])$ and $\nu_i = \sigma_i/\tau_i$. Disjoint restrictions are a special case; fractional restrictions of atoms are also allowed. Choose

$$0 < r_i \leq 1/4, \qquad \sum_i \tau_i r_i \leq \rho.$$

Within component i, use the half-prior subclass

$$X_{r_i}(\nu_i) = \{0 \leq q \leq 1/2 : \ \textstyle\int q \, d\nu_i \leq r_i\}.$$

The original population lower bound is valid on this subclass of the full rarity class.

Theorem C.7 (Additive localization). *Suppose that, for each i, every capped bounded observable one-unit output with uniform bias at most $b_i > 0$ over $X_{r_i}(\nu_i)$ satisfies*

$$\left(\sup_q \operatorname{Var}_q Y\right) \mathbb{E}_{i,0} T \geq J_i > 0.$$

Then, for every $B > 0$,

$$\boxed{R_{\mu,\rho}(B) \geq \frac{1}{128} \sum_{i=1}^{m} \tau_i \min\{b_i, \sqrt{J_i/B}\}.} \tag{72}$$

The same total budget B occurs in every summand. The component label is chosen before the first observation and retained on every revisit; its conditional magnitude and sign remain hidden.

Proof. Theorem C.3 applies within every component. Set

$$\delta_i = \frac{1}{32} \min\{b_i, \sqrt{J_i/B}\}.$$

It supplies a pair $q_{i,-}, q_{i,+} \in X_{r_i}(\nu_i)$, oriented to have conditional target separation at least δ_i, such that simultaneously for every capped local policy A,

$$h(P^A_{i,-}, P^A_{i,+}) \leq 42 \frac{\delta_i^2}{J_i} \mathbb{E}_{i,0} T_A \leq \frac{42}{1024B} \mathbb{E}_{i,0} T_A. \tag{73}$$

Consider a stronger experiment that can select a component at no cost and then receive a genuine coin with magnitude law ν_i. Its component label is visible, while its magnitude and sign remain hidden. The algorithm can adaptively select, revisit, and interleave these coins. Under the conditional midpoint prior, $q \leq 1/2$, so the midpoint law dominates half the null law. Lemma C.2 therefore bounds terminal squared Hellinger by $84/1024 < 1/8$: the local coefficient in (73) is common to all components, and the summed conditional midpoint costs count each purchased observation once. Increasing capped prefixes handles unbounded stopping. The total target separation is at least $\sum_i \tau_i \delta_i$.

Every original algorithm can be simulated in this stronger experiment. Privately choose the component label with probabilities τ_i whenever it creates a fresh coin. On the remaining probability mass, simulate the known all-negative residual law for free. Keep the chosen label when the coin is revisited. For fractional submeasures, integrating this private mixture label gives the original admissible parameters

$$q_\pm(d) = \sum_i \frac{d\sigma_i}{d\mu}(d) q_{i,\pm}(d), \qquad 0 \le q_\pm \le 1/2, \qquad \int q_\pm \, d\mu \le \sum_i \tau_i r_i \le \rho.$$

The number of genuine oracle observations does not exceed the original number. Thus the oracle lower bound applies to the original experiment. The Hellinger bound gives TV at most $1/2$, and midpoint testing gives RMS at least one quarter of the target separation. This is (72). □

For a direct explicit input, suppose component i is supported in $(0, a_i]$, $a_i \le 1/4$, and select a finite separated family of small-gap annuli. Let Γ_i° and $w_{*,i}$ denote its conditional cost entropy and minimum band mass. Proposition B.4 gives

$$b_i = r_i/64, \qquad J_i = c_{\mathrm{ann}} \min\{r_i \Gamma_i^\circ, w_{*,i}/a_i^2\}, \qquad c_{\mathrm{ann}} = \frac{e^{-28}}{2{,}000{,}000}.$$

Write the corresponding unnormalized quantities with respect to σ_i as $\Gamma_{\sigma_i}^\circ = \tau_i \Gamma_i^\circ$ and $w_{*,\sigma_i} = \tau_i w_{*,i}$. The conversion uses the exact identity, with $\theta = \tau r$,

$$\tau^2 \min\{r\Gamma^\circ, w_*/a^2\} = \min\{\theta \Gamma_\sigma^\circ, \tau w_{*,\sigma}/a^2\}.$$

In particular the second branch retains the factor τ. Consequently,

$$R_{\mu,\rho}(B) \gtrsim \sum_i \min\left\{\tau_i r_i, \sqrt{\frac{\min\{\tau_i r_i \Gamma_{\sigma_i}^\circ, \tau_i w_{*,\sigma_i}/a_i^2\}}{B}}\right\}. \tag{74}$$

Supremizing this expression over finite decompositions and rarity allocations gives an explicit lower functional. The bias-indexed form (72) also permits each component's accuracy scale to be chosen separately. Section 4 applies this expression to components selected by the entropy–tail optimizer.

C.7 A universal risk floor

Lemma C.8 (Bernoulli-label floor)**.** *For every magnitude law, $0 < \rho \le 1/4$, and $B > 0$,*

$$R_{\mu,\rho}(B) \ge \frac{1}{16} \min\{\rho, \sqrt{\rho/B}\}.$$

Proof. Restrict to constant sign allocations. Give an oracle the exact sign of a unit when its first observation is requested, for one charged observation. It can sample $D \sim \mu$ independently and simulate all future flips for free, so every original algorithm can be simulated with no greater work. The informative labels are iid Bernoulli with the unknown constant prior.

Take $p_0 = \rho/2$ and $p_1 = p_0 + \delta$, where $\delta = \min\{\rho/2, \sqrt{\rho/B}/4\}$. Both are admissible and $p_1(1-p_1) \ge 3\rho/8$. The stopped KL chain rule, first at bounded prefixes and then by monotone convergence, gives

$$\mathrm{KL}(P_0, P_1) \le B \ \mathrm{kl}(p_0, p_1) \le \frac{8B\delta^2}{3\rho} \le \frac{1}{6}.$$

Thus $\mathrm{TV}(P_0, P_1) \le 1/2$. Thresholding any estimator at the target midpoint implies worst RMS at least $\delta/4$. Finally $\delta/4 = \min\{\rho/8, \sqrt{\rho/B}/16\} \ge \min\{\rho, \sqrt{\rho/B}\}/16$. □

D Representation and realization of the stopping flow

This section justifies the normalized flow cone and robust program in (81)–(82). Triangular encodings of stopped coin experiments have direct antecedents in Lee–Valiant [3]; occupation-measure formulations of optimal stopping are also classical [9]. The proofs below give the representation and estimator realization needed for this particular robust program.

D.1 The support function of the rarity class

For bounded measurable $f : (0, 1/2] \to \mathbb{R}$, put

$$Q_{\mu,\rho}(f) = \sup_{q \in \mathcal{Q}_\rho} \int q(d) f(d)\, \mu(dd).$$

Lemma D.1 (A fractional upper-tail formula)**.** *The support function satisfies the attained identity*

$$Q_{\mu,\rho}(f) = \min_{\ell \geq 0} \left\{ \rho\ell + \int (f - \ell)_+ \, d\mu \right\}. \tag{75}$$

The minimum can be restricted to $[0, \|f_+\|_\infty]$*. The supremum on the left is attained by selecting the largest positive values of* f*, allowing a fractional selection on the cutoff level set.*

Proof. For every $0 \leq q \leq 1$ and $\ell \geq 0$, $qf \leq q\ell + (f - \ell)_+$. Integration and $\int q \, d\mu \leq \rho$ prove the upper bound by the right side.

If $\mu(f > 0) \leq \rho$, take $q = \mathbf{1}_{\{f>0\}}$ and $\ell = 0$. Otherwise choose a positive cutoff ℓ satisfying

$$\mu(f > \ell) \leq \rho \leq \mu(f \geq \ell).$$

Such a cutoff exists by the monotonicity and one-sided continuity of the tail distribution. Take $q = 1$ on $\{f > \ell\}$, $q = 0$ on $\{f < \ell\}$, and choose a constant fraction in $[0, 1]$ on $\{f = \ell\}$ so that its total mass is ρ. The resulting value is

$$\int_{\{f>\ell\}} f \, d\mu + \ell[\rho - \mu(f > \ell)] = \rho\ell + \int (f - \ell)_+ \, d\mu.$$

This proves equality and attainment, including atoms of μ or atoms in the distribution of f. Cutoffs larger than $\|f_+\|_\infty$ cannot improve on the endpoint. □

For an array $u = (u_{nk})$, define

$$L_u^{\pm}(d) = \sum_{n,k} u_{nk} \binom{n}{k} (1/2 \pm d)^k (1/2 \mp d)^{n-k}, \qquad I_u^- = \int L_u^- \, d\mu, \qquad F_u = L_u^+ - L_u^-.$$

The root or final-depth terms are included only when the corresponding array permits them. The convention $0^0 = 1$ covers the deterministic Bernoulli endpoints. By the definition of $m(q)$,

$$\sup_{q \in \mathcal{Q}_\rho} \langle m(q), u \rangle = I_u^- + Q_{\mu,\rho}(F_u). \tag{76}$$

In particular, the two bias suprema in (82) are

$$c_0 + I_z^- + Q_{\mu,\rho}(F_z - 1), \qquad -c_0 - I_z^- + Q_{\mu,\rho}(1 - F_z).$$

The moment and work suprema use (76) with $u = t$ and $u = c$. Thus the robust program requires only integration against the known magnitude law and a scalar cutoff optimization.

D.2 Arbitrary seeds and ordered histories aggregate to the cone

Proposition D.2 (Normalized prefix aggregation)**.** *Every capped one-unit policy with* $1 \le T \le N$*, arbitrary independent auxiliary randomization, and bounded terminal output has normalized arrays in the cone* (81) *with root intensity one. Its mean, raw second moment, and expected work under any* q *are respectively*

$$\langle m(q), z\rangle, \qquad \langle m(q), t\rangle, \qquad \langle m(q), c\rangle.$$

Multiplying the four arrays by $\lambda \ge 0$ *gives the corresponding intensity-*λ *representation. The stopping representation alone requires no output or bounded-output assumption.*

Proof. Place all auxiliary randomization in one seed r, drawn from a law ν independent of the latent unit and its observations. Once the seed is fixed, all decisions and outputs are deterministic functions of the ordered observed prefix.

For each ordered binary prefix x of length n, let A_x be the set of seeds for which the policy reaches x and stops there; let C_x be the set for which it reaches x and continues. These sets are disjoint. For $n \ge 1$ their union is exactly C_{x^-}, where x^- is the prefix with its last symbol removed. Write

$$\alpha_x = \nu(A_x), \quad \gamma_x = \nu(C_x), \quad \zeta_x = \int_{A_x} Y(r,x)\,\nu(dr), \quad \tau_x = \int_{A_x} Y(r,x)^2\,\nu(dr).$$

At the root, $\gamma_\varnothing = 1$ and $\alpha_\varnothing = 0$; at depth N, $\gamma_x = 0$.

For $s = (n,k)$, sum these quantities over all prefixes with k heads and divide by $\binom{n}{k}$:

$$a_{nk} = \frac{\sum_{|x|=n,\,\#1(x)=k} \alpha_x}{\binom{n}{k}}, \qquad c_{nk} = \frac{\sum_{|x|=n,\,\#1(x)=k} \gamma_x}{\binom{n}{k}},$$

and analogously define z_{nk} and t_{nk} from ζ_x, τ_x. Aggregating $\alpha_x + \gamma_x = \gamma_{x^-}$ gives

$$a_{nk} + c_{nk} = \frac{n-k}{n} c_{n-1,k} + \frac{k}{n} c_{n-1,k-1}.$$

Cauchy–Schwarz on the disjoint union of the prefix-indexed stopping-seed sets gives $z_{nk}^2 \le a_{nk} t_{nk}$. Hence the arrays belong to the stated cone.

Conditional on Bernoulli mean p, every ordered prefix in this count state has the same probability $p^k(1-p)^{n-k}$. Summing its output integrals therefore gives the stated moment identities after mixing over the latent state. Expected work is the sum of the probabilities of continuing at all prefixes, including the root: each continuation requests exactly one charged observation. This proves the work identity and accounts for the first observation.

Multiplication by λ preserves the linear flow identities and the homogeneous perspective inequalities. The argument also applies to outputs with finite first and second seed aggregates. For the bounded outputs used in finite constructions, these are automatic. □

The normalized arrays encode probabilities over the independent seed for fixed observation prefixes. They are not the unconditional probabilities of reaching a count state. This distinction is the reason the Pascal coefficients in (81) are $(n-k)/n$ and k/n.

D.3 Every feasible point is an observable estimator

Proposition D.3 (Compound-Poisson realization)**.** *Every feasible point of* (82) *constructs an observable population estimator with uniform expected work at most* B *and worst RMS error at most* $\sqrt{b^2+v}$*. In particular,*

$$R_{\mu,\rho}(B) \le P_N(B).$$

Proof. Suppose first that $\lambda > 0$. Draw a Poisson number of independent genuine units with mean λ. Each unit first requests one observation. At an observed state (n, k), stop with probability $a_{nk}/(a_{nk} + c_{nk})$ and continue otherwise. At a stopping state with $a_{nk} > 0$, return z_{nk}/a_{nk}. Unreachable states may have arbitrary actions, and outputs at zero stopping flow can be set to zero. The constraint $c_{Nk} = 0$ enforces the cap.

For a fixed Bernoulli mean p, the expected number of units reaching (n, k) is

$$(a_{nk} + c_{nk})\binom{n}{k}p^k(1-p)^{n-k}.$$

This follows by induction from the normalized Pascal recursion. For example, a continuation at $(n-1, k)$ followed by a zero contributes $c_{n-1,k}\binom{n-1}{k}p^k(1-p)^{n-k}$, and the other parent contributes the analogous term with $k-1$. Their sum is exactly the displayed expression. Multiplying by the stopping or continuation probability gives the same formula with a_{nk} or c_{nk} alone.

Return c_0 plus the sum of all terminal unit outputs. Independence of the genuine units and the compound-Poisson moment identities give

$$\mathbb{E}_q\widehat{\theta} = c_0 + \langle m(q), z\rangle, \tag{77}$$

$$\mathrm{Var}_q(\widehat{\theta}) = \left\langle m(q), \frac{z^2}{a}\right\rangle \le \langle m(q), t\rangle, \tag{78}$$

$$\mathbb{E}_q T = \langle m(q), c\rangle. \tag{79}$$

Set $z_{nk}^2/a_{nk} = 0$ where $a_{nk} = 0$; the perspective constraint then also gives $z_{nk} = 0$. Every sum is finite, and each nonzero terminal output is a finite real number. The root contribution to (79) is λ, since $m_{00}(q) = 1$. The work constraint therefore charges every first observation and implies $\lambda \le B$.

The bias, moment, and work inequalities in the program prove the claimed uniform guarantees. When $\lambda = 0$, the flow recursion forces $a = c = 0$ and the perspective constraint forces $z = 0$. The deterministic estimator c_0 realizes the required bound; any unused nonnegative moment slack is harmless. □

Allowing zero-observation marks would not improve the attainable Poisson guarantee. Indeed, the event that a one-unit procedure stops before its first observation depends only on its independent seed. Poisson thinning separates those marks from the active units. Replace their total random contribution by its deterministic expectation. This preserves every population mean and the work, while removing their nonnegative variance contribution. The remaining active intensity, together with the adjusted offset c_0, has precisely the root convention in (81).

D.4 Finite-support laws give finite second-order cone programs

Suppose $\mu = \sum_{i=1}^m w_i\delta_{d_i}$ is supplied by its positive masses and distinct positive magnitudes. For each of the four support functions in the two bias, moment, and work constraints, introduce one cutoff $\ell \ge 0$ and nonnegative epigraph variables r_i with

$$r_i \ge f(d_i) - \ell.$$

The corresponding robust term is represented by $\rho\ell + \sum_i w_i r_i$, by Lemma D.1. All $f(d_i)$ are affine functions of the relevant flow or output array. Thus these epigraph constraints are linear.

The perspective constraints are second-order cones:

$$z_{nk}^2 \le a_{nk}t_{nk} \quad\Longleftrightarrow\quad \|(2z_{nk}, a_{nk} - t_{nk})\|_2 \le a_{nk} + t_{nk}, \qquad a_{nk}, t_{nk} \ge 0.$$

One additional cone represents an epigraph variable $u \geq b^2$, for example $\|(2b, u-1)\|_2 \leq u+1$, and the objective is $u+v$.

Consequently the finite-support version has $O(N^2+m)$ scalar variables and cone or linear constraints, with a dense coefficient description of size $O(mN^2)$ after the Bernoulli kernels are evaluated. This is a dimension count at a specified cap, without a uniform claim about the sufficient cap, numerical conditioning, or bit complexity.

E An exact dual for the robust stopping program

Triangular stopping representations and their adaptive comparison have close antecedents in Lee–Valiant [3]. Here the sign allocation ranges over the full convex rarity class, and bias, variance, and work are robust constraints. We derive the exact four-prior dual of this program and absorb its variance and work references into the endpoints. Appendix D proves the stopping-flow representation and its observable realization in full.

Throughout this section, μ is any probability measure on $(0, 1/2]$, $0 < \rho \leq 1/4$, $B > 0$, and $N \geq 1$ is an integer. Put

$$\mathcal{Q}_\rho = \{q \in L^\infty(\mu) : 0 \leq q \leq 1,\ \theta(q) = \textstyle\int q\, d\mu \leq \rho\}.$$

For a state $s = (n, k)$, $0 \leq k \leq n \leq N$, define

$$m_{nk}(q) = \int \binom{n}{k} \big[(1-q(d))(1/2-d)^k(1/2+d)^{n-k} + q(d)(1/2+d)^k(1/2-d)^{n-k}\big]\, \mu(dd). \tag{80}$$

In particular $m_{00}(q) = 1$. These are probabilities of count states at fixed depth, before imposing a stopping rule.

E.1 The primal and its four-prior dual

For clarity, the finite program being dualized is recalled in compact form. Its nonnegative stopping and continuation arrays a, c, real output array z, and nonnegative moment array t obey

$$\begin{aligned} &a_{00} = z_{00} = t_{00} = 0, \qquad c_{00} = \lambda \geq 0, \qquad c_{Nk} = 0, \\ &a_{nk} + c_{nk} = \frac{n-k}{n} c_{n-1,k} + \frac{k}{n} c_{n-1,k-1}, \qquad 1 \leq n \leq N, \\ &z_{nk}^2 \leq a_{nk} t_{nk}. \end{aligned} \tag{81}$$

Missing indices are zero. These are normalized prefix-weight flows, not unconditional state probabilities. Write $\mathcal{C}_N$ for this closed convex cone. The squared program value is

$$\begin{aligned} P_N(B)^2 = \min\ & b^2 + v \quad \text{over } (a, c, z, t, \lambda) \in \mathcal{C}_N,\ c_0 \in \mathbb{R},\ b, v \geq 0, \\ \text{subject to }\ & \sup_{q \in \mathcal{Q}_\rho} \big|c_0 + \langle m(q), z\rangle - \theta(q)\big| \leq b, \\ & \sup_q \langle m(q), t\rangle \leq v, \qquad \sup_q \langle m(q), c\rangle \leq B. \end{aligned} \tag{82}$$

Stopping sums exclude the root and continuation sums exclude depth N. Every feasible point realizes a Poisson estimator with uniform RMS at most $\sqrt{b^2+v}$ and expected work at most B (Proposition D.3).

Use the closed perspective

$$\mathfrak{r}(x,y)=\begin{cases} x^2/y, & y>0,\\ 0, & x=y=0,\\ +\infty, & y=0, x\neq 0. \end{cases}$$

Given four priors $q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho$, put $d_s = m_s(q_H) - m_s(q_L)$, $v_s = m_s(q_v)$, $w_s = m_s(q_w)$, and define

$$K_N(q_L,q_H;q_v,q_w)=\sup_{A:\,1\le T_A\le N}\frac{\sum_s a_s^A \mathfrak{r}(d_s,v_s)}{\sum_s c_s^A w_s}. \tag{83}$$

Here A is a one-unit policy that always requests its first observation; its normalized flow has root intensity one. A zero stopping coefficient contributes zero even when the perspective is infinite. The denominator is $\mathbb{E}_{q_w} T_A \in [1,N]$.

For a signed measure $P_H - P_L$, write

$$\mathcal{I}(P_H,P_L;P_v)=\int\left(\frac{d(P_H-P_L)}{dP_v}\right)^2 dP_v,$$

with value infinity unless the signed measure is absolutely continuous. The numerator in (83) equals $\mathcal{I}(P_H^A,P_L^A;P_v^A)$. Include the independent seed in the transcript: its likelihood ratio depends only on the terminal count state, so this identity covers order-dependent stopping and randomized policies as well.

Theorem E.1 (Exact four-prior dual)**.** *The minimum in (82) is attained, and*

$$\boxed{P_N(B)^2=\max_{q_L,q_H,q_v,q_w\in\mathcal{Q}_\rho}\frac{[\theta(q_H)-\theta(q_L)]_+^2}{4+BK_N(q_L,q_H;q_v,q_w)}.} \tag{84}$$

The ratio is zero when $K_N=\infty$. The maximum is attained with positive value, positive target separation, and finite K_N. Swapping the endpoints allows the positive part to be replaced by an absolute value.

Proof. Compactness and constraint qualification. The finite moment image $\{(\theta(q),m(q)) : q\in\mathcal{Q}_\rho\}$ is compact and convex: $\mathcal{Q}_\rho$ is weak-star compact, and every coordinate is a bounded linear functional of q. All robust functions in the program are thus finite continuous convex functions.

The zero-flow output $\rho/2$ gives objective $\rho^2/4$. On this objective sublevel, $b\le\rho/2$, $v\le\rho^2/4$, and the root work charge gives $\lambda\le B$. The normalized recursion implies $0\le a_s,c_s\le B$. For the constant prior $q_*=\rho/2$, $m_s(q_*)>0$ unless state s is impossible under every prior. At every nonnull stopping state,

$$t_s\le\frac{\rho^2}{4m_s(q_*)},\qquad |z_s|\le\sqrt{Bt_s}.$$

At universally null states set $z_s=t_s=0$, without changing any mean, moment, or cost. The bias constraint at $q=0$ then bounds c_0. Consequently this feasible sublevel, with null output coordinates removed, is compact. The primal minimum is attained.

There is also a strict relative-interior feasible point. Choose $0<\lambda<B/4$, continue with probability $1/2$ at each nonterminal observed state, and stop at depth N. All free stopping and continuation flows are positive and expected work is less than $2\lambda<B$. Set $z=0$, take all free $t_s>0$, and choose sufficiently large b,v. Every perspective inequality and all four robust inequalities are strict. Finite-dimensional convex Slater duality therefore gives no duality gap, attained inequality multipliers, and KKT conditions at a primal optimum.

Reduction to four priors. Split the absolute bias constraint into its two one-sided inequalities. Each robust function is a maximum of affine functions over a compact convex set. A subgradient is a convex combination of active affine gradients. Since $q \mapsto (\theta(q), m(q))$ is affine and $\mathcal{Q}_\rho$ is convex, this combination is represented by one active prior. Thus the two bias subgradients, the variance subgradient, and the work subgradient may be represented respectively by q_L, q_H, q_v, q_w.

Let their nonnegative multipliers be $\alpha_L, \alpha_H, \beta, \eta$. The free offset forces $\alpha_L = \alpha_H = \alpha$. Minimizing $b^2 - 2\alpha b$ over $b \geq 0$ gives $-\alpha^2$, and minimizing $(1-\beta)v$ over $v \geq 0$ requires $\beta \leq 1$. The remaining expression on the flow cone is

$$\beta\langle m(q_v), t\rangle - \alpha\langle m(q_H) - m(q_L), z\rangle + \eta\langle m(q_w), c\rangle. \tag{85}$$

Homogeneity requires this expression to be nonnegative everywhere on $\mathcal{C}_N$; otherwise scaling a negative point gives infimum minus infinity. When it is nonnegative its infimum is zero. The dual objective is

$$\alpha[\theta(q_H) - \theta(q_L)] - \alpha^2 - \eta B. \tag{86}$$

We may take $\beta = 1$. Indeed, for any $q_0 \in \mathcal{Q}_\rho$, replace $\beta m(q_v)$ by

$$m\big(\beta q_v + (1-\beta)q_0\big) = \beta m(q_v) + (1-\beta)m(q_0).$$

The additional coefficient on t is nonnegative, so cone feasibility is preserved while the dual objective is unchanged. This also preserves attainment.

Eliminating outputs and the work price. For $\alpha > 0$, the perspective constraint gives, at fixed stopping flow,

$$\inf_{z_s^2 \leq a_s t_s} \{v_s t_s - \alpha d_s z_s\} = -\frac{\alpha^2 a_s}{4}\mathfrak{r}(d_s, v_s).$$

For positive v_s this is completion of the square. If $a_s = 0$ the contribution is zero; if $a_s > 0$, $v_s = 0$, and $d_s \neq 0$, the infimum is minus infinity. The remaining zero case has value zero. Every nonzero stopping/continuation flow scales to root intensity one. At $\lambda = 0$, $a = c = z = 0$ and surplus t has a nonnegative contribution. Thus nonnegativity of (85) is exactly

$$\eta \geq \alpha^2 K_N/4.$$

For finite K_N, optimizing (86) yields

$$\sup_{\alpha \geq 0}\Big\{\alpha\Delta\theta - \alpha^2(1 + BK_N/4)\Big\} = \frac{(\Delta\theta)_+^2}{4 + BK_N}.$$

When $K_N = \infty$, only the zero-value choice $\alpha = \eta = 0$ remains. Weak duality applies to every quadruple, and the attained KKT dual above supplies an optimal quadruple. This proves the identity.

Finally take $q_L = 0$ and $q_H = q_v = \rho$ constant. The divergence of the latent sign contrast relative to q_v is $\rho/(1-\rho)$. Data processing bounds every stopped-transcript numerator by this quantity, and work is at least one. Thus, for any q_w,

$$P_N(B)^2 \geq \frac{\rho^2}{4 + B\rho/(1-\rho)} > 0. \tag{87}$$

An attained optimal dual therefore has $\alpha > 0$, finite K_N, and positive target separation, as asserted. □

E.2 A Bellman recursion with explicit null-state conventions

Divide each coefficient in (80) by $\binom{n}{k}$ and write $p_q(n,k)$ for the probability of one particular ordered prefix with that count. These coefficients satisfy

$$p_q(n,k) = p_q(n+1,k) + p_q(n+1,k+1).$$

In this subsection let $d_s = p_H(s) - p_L(s)$, $v_s = p_v(s)$, $w_s = p_w(s)$, and $r_s = \mathfrak{r}(d_s, v_s)$ denote the corresponding *prefix* coefficients. At the root $v_0 = w_0 = 1$ and $d_0 = r_0 = 0$.

Proposition E.2 (Bellman representation)**.** *Suppose $K_N < \infty$. For $\kappa \geq 0$, set*

$$\begin{aligned} U_{Nk}(\kappa) &= r_{Nk}, \\ U_{nk}(\kappa) &= \max\{r_{nk}, U_{n+1,k}(\kappa) + U_{n+1,k+1}(\kappa) - \kappa w_{nk}\}, \quad 1 \leq n < N. \end{aligned} \tag{88}$$

Then

$$\Phi_N(\kappa) := U_{10}(\kappa) + U_{11}(\kappa) - \kappa = \max_{1 \leq T_A \leq N} \{\mathcal{I}(P_H^A, P_L^A; P_v^A) - \kappa \mathbb{E}_{q_w} T_A\}. \tag{89}$$

It is finite, continuous, piecewise affine, and strictly decreasing, and its unique nonnegative zero is K_N. A deterministic count-state policy attains the maximum at every price. If some prefix has $v_s = 0$ and $d_s \neq 0$, then $K_N = \infty$ instead.

Proof. At any observed prefix, stopping earns r_s and continuation earns the two child values minus the work charge κw_s. Backward induction therefore proves (88). Root activation is compulsory, which gives the final charge κ. All coefficients depend only on the count, so the maximizing choice may be deterministic and depend only on (n,k). Randomized or order-dependent choices cannot improve it. There are finitely many such deterministic policies, each with work in $[1,N]$. Their affine price values have slopes in $[-N,-1]$, proving the continuity and strict monotonicity claims. The zero-price value is nonnegative, the value tends to minus infinity, and the zero is precisely the maximal information-to-work ratio. If a prefix has zero reference likelihood and nonzero contrast, the policy stopping at its depth has infinite information. □

The recursion also admits an information-increment form. Put

$$j_{nk} = r_{n+1,k} + r_{n+1,k+1} - r_{nk} \geq 0. \tag{90}$$

Nonnegativity is Cauchy–Schwarz. At a zero-reference state, finite K_N forces both its contrast and its children's contrasts to vanish, so $j_s = 0$ there. If b_s^A is the unnormalized continuation-prefix flow, telescoping the flow identities gives

$$\mathcal{I}(P_H^A, P_L^A; P_v^A) = \sum_{n<N,k} b_s^A j_s, \qquad \mathbb{E}_{q_w} T_A = \sum_{n<N,k} b_s^A w_s. \tag{91}$$

Subtracting r_s from the Bellman values gives terminal values $V_N = 0$ and

$$V_s(\kappa) = \max\{0, j_s - \kappa w_s + V_{s0}(\kappa) + V_{s1}(\kappa)\}, \qquad \Phi_N(\kappa) = j_0 - \kappa + V_{10}(\kappa) + V_{11}(\kappa).$$

Thus the ratio accounts for both information gained at a state and the work needed to reach it. The pointwise inequalities $j_s \leq \kappa w_s$ are sufficient, though generally not necessary, for $K_N \leq \kappa$.

When q_v dominates q_w on all relevant prefixes, the posterior form is

$$\Phi_N(\kappa) = \sup_{1 \leq T \leq N} \mathbb{E}_{q_v} \left[L_T^2 - \kappa \sum_{n<T} W_n \right], \qquad L_n = d_n/v_n, \quad W_n = w_n/v_n.$$

The process L_n is a signed-likelihood martingale, initially zero. The unnormalized Bellman recursion itself requires no such domination assumption, and is the appropriate form at degenerate states.

F A midpoint modulus and a sharper adaptive comparison

Define the two-prior quantity

$$D_N(B)^2 = \sup_{q_-,q_+\in\mathcal{Q}_\rho} \frac{|\theta(q_+)-\theta(q_-)|^2}{4+BK_N(q_-,q_+;\bar q,\bar q)}, \qquad \bar q=(q_-+q_+)/2. \tag{92}$$

Theorem F.1 (Absorbing the two references)**.** *For every law, budget, and finite cap in the stated range,*

$$D_N(B) \le P_N(B) \le 3D_N(B). \tag{93}$$

Proof. The first inequality restricts the four-prior dual. For the second, start with any quadruple of finite K_N and set

$$q'_- = (q_L+q_v+q_w)/3, \qquad q'_+ = (q_H+q_v+q_w)/3.$$

Their midpoint dominates q_v and q_w in the following precise experiment sense: for every policy, $P^A_{\bar q'} \ge P^A_{q_v}/3$ and $\mathbb{E}_{\bar q'}T_A \ge \mathbb{E}_{q_w}T_A/3$. The signed transcript contrast is the old contrast divided by three. Consequently its information numerator is at most one third of the old numerator, and

$$K_N(q'_-,q'_+;\bar q',\bar q') \le K_N(q_L,q_H;q_v,q_w).$$

The target gap is divided by three. Substitution in (92) and then (84) proves the result. □

For midpoint references, the numerator is twice triangular discrimination. With $h(P,Q) = 1-\mathrm{Aff}(P,Q)$,

$$4h(P_+,P_-) \le 2\int \frac{(p_+-p_-)^2}{p_++p_-} = \mathcal{I}(P_+,P_-;(P_++P_-)/2) \le 8h(P_+,P_-). \tag{94}$$

These inequalities follow by factoring $(p_+-p_-)^2 = (\sqrt{p_+}-\sqrt{p_-})^2(\sqrt{p_+}+\sqrt{p_-})^2$ and bounding the second factor divided by p_++p_- between one and two. Hence the midpoint modulus is also a Hellinger-information-per-work modulus, up to universal constants. It still involves an optimal local stopping rule; it is not a closed tail-moment or entropy formula.

Lemma F.2 (Squared-loss testing by affinity)**.** *If two terminal laws have target gap Δ, every estimator has worst RMS at least $\Delta\,\mathrm{Aff}(P_-,P_+)/2$.*

Proof. The maximum MSE is at least the average MSE for equally likely endpoints. The minimum average MSE is

$$\frac{\Delta^2}{2}\int \frac{p_-p_+}{p_-+p_+}.$$

Cauchy–Schwarz gives $\mathrm{Aff}(P_-,P_+)^2 \le 2\int p_-p_+/(p_-+p_+)$, proving the claim. □

Let $R^{\mathrm{tot}}_N(B)$ denote population minimax RMS with at most N total observations and uniform expected work at most B. This total cap is different from the per-unit cap used in P_N.

Theorem F.3 (A direct adaptive comparison)**.**

$$R^{\mathrm{tot}}_N(B) \ge \frac{3}{8}D_N(B), \qquad P_N(B) \le 8R^{\mathrm{tot}}_N(B). \tag{95}$$

Proof. Fix a pair in (92), with target gap Δ and information ratio K. Interpolate its endpoints toward their midpoint by $s \in [0,1]$. The midpoint stays fixed, the target gap becomes $s\Delta$, and the information ratio becomes exactly s^2K.

For every induced local policy of a cap-N population algorithm, (94) gives

$$h(P_-^A, P_+^A) \le \frac{s^2K}{4}\mathbb{E}_{\bar{q}}T_A.$$

This inequality also covers policies that sometimes take zero observations. Their zero-work event depends only on the seed and has zero signed contrast; conditioning on activation gives a root-active policy, and both information and expected work scale by the activation probability. Suppressing the seed can only decrease information.

Apply Lemma C.2. It realizes each hybrid as a one-genuine-unit policy, simulating other units from independent midpoint tapes in its auxiliary seed. The hybrid global transcript is a function of that local transcript, so data processing applies even if the seed reveals the other tapes. Summing the midpoint local work counts every population observation exactly once. Therefore

$$h_{\mathrm{pop}} \le s^2BK/4.$$

If $K = 0$ take $s = 1$; otherwise take $s = \min\{1, (BK)^{-1/2}\}$. Then $h_{\mathrm{pop}} \le 1/4$. Lemma F.2 yields

$$R_N^{\mathrm{tot}}(B) \ge \frac{3}{8}s\Delta \ge \frac{3}{8}\frac{\Delta}{\sqrt{4+BK}}.$$

Taking the pair supremum proves the first claim; the second follows from Theorem F.1. □

Corollary F.4 (An improved finite characterization)**.** *For every integer $N \ge 1$,*

$$\left(\frac{P_N(B)}{8} - \frac{\rho}{2}\sqrt{\frac{B}{N+1}}\right)_+ \le R_{\mu,\rho}(B) \le P_N(B). \tag{96}$$

At the explicit cap

$$N_* = \left\lceil 64\max\{B, \rho B^2\}\right\rceil,$$

this gives

$$\boxed{R_{\mu,\rho}(B) \le P_{\mu,\rho,N_*}(B) \le 16R_{\mu,\rho}(B).} \tag{97}$$

Proof. Clip any estimator to $[0,\rho]$ and, before requesting observation $N+1$, returns $\rho/2$. Markov's inequality and the L^2 triangle inequality give

$$R_N^{\mathrm{tot}}(B) \le R_{\mu,\rho}(B) + \frac{\rho}{2}\sqrt{\frac{B}{N+1}}.$$

Combining this with Theorem F.3 and the constructive flow upper bound proves (96). Lemma C.8 gives

$$R_{\mu,\rho}(B) \ge \frac{1}{16}\min\{\rho, \sqrt{\rho/B}\}.$$

At $N = N_*$ the truncation remainder is at most that floor. Thus $P_{N_*}(B) \le 8(R+R) = 16R$. □

The program remains a uniform risk certificate rather than an exact formula for minimax risk, and its finite dimension does not imply an efficient algorithm at every budget. Theorem 3.2 supplies a complementary additive expression involving only the magnitude law, with its matching lower proof in Section 4. Its conservative scalar constants are distinct from this program factor.

Remark F.5 (Scope of the factor-sixteen guarantee). The comparison uses the sufficient cap in Corollary F.4, or a cap that passes the verified genie stopping condition in Section G. It is not a comparison for every fixed cap. In particular, an ordinary fixed-cap dual alone is not a lower certificate for unrestricted risk. With verified bounds $L \le G_N$ and $P_N \le U$, the orientation is

$$L \le G_N \le P_\infty \le P_N \le U, \qquad L/8 \le R \le U.$$

Thus $U \le 2L$ implies $U \le 16R$.

G Finite certificates for the unrestricted experiment

The universal cap in Corollary F.4 is a sufficient theoretical cap. This section gives a different computational route: upper and lower programs at a chosen cap, with an a posteriori stopping criterion. The lower program strengthens the one-unit experiment by revealing its latent magnitude and sign if it reaches that cap. This reveal-at-cap device has a direct precedent in Lee–Valiant [3, Section 5.2]; the contribution here is its convergent robust dual bracket and exact stopping certificate. The extra information is used only in a lower-bound argument. The upper estimator continues to observe ordinary Bernoulli samples alone.

G.1 The infinite-cap dual and its compactness argument

Define

$$P_\infty(B) = \inf_{N\ge 1} P_N(B), \qquad K_\infty(q_L, q_H; q_v, q_w) = \sup_{N\ge 1} K_N(q_L, q_H; q_v, q_w).$$

The latter definition uses only deterministically capped local policies. No assertion about a general unbounded policy is needed for the following interchange.

Lemma G.1 (The exact dual persists in the limit)**.** *For every law and every $B > 0$,*

$$P_\infty(B)^2 = \max_{q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_\infty(q_L, q_H; q_v, q_w)}. \tag{98}$$

The maximum is attained and is positive. If D_∞ is the corresponding midpoint-reference modulus, then

$$D_\infty \le P_\infty \le 3D_\infty, \qquad R(B) \le P_\infty(B) \le 8R(B). \tag{99}$$

Proof. The space $\mathcal{Q}_\rho^4$ is weak-star compact. For each fixed cap, K_N is lower semicontinuous: every fixed policy has a lower semicontinuous closed-perspective numerator and a positive continuous work denominator, and taking their supremum preserves lower semicontinuity. Its dual ratio $f_N = (\Delta\theta)_+^2/(4 + BK_N)$ is upper semicontinuous, with value zero at an infinite denominator. Indeed, the numerator is continuous and bounded, and lower semicontinuity forces a diverging denominator when approaching a point with $K_N = \infty$. The functions f_N decrease with N. If $M = \inf_N \max f_N$, the nonempty compact sets $\{f_N \ge M\}$ are nested. Their intersection gives a quadruple with $\inf_N f_N \ge M$. The reverse inequality is immediate. This proves (98) and attainment. Positivity follows from (87).

The equal-third reference absorption in Theorem F.1 holds simultaneously at every cap, hence also for K_∞. It proves the first two comparisons. For a fixed midpoint pair, the hybrid Hellinger bound holds at every finite population cap, with the same coefficient K_∞. Increasing capped prefixes extends the affinity bound to the full terminal experiment of any finite-expected-work algorithm. The testing and interpolation argument of Theorem F.3 therefore gives $R \ge 3D_\infty/8$. The constructive inequality $R \le P_N$ holds for every N. These facts prove (99). □

G.2 Revealing the latent state at the cap

Fix four priors and write $\delta = q_H - q_L$. Continue to use the prefix coefficients from Proposition E.2. For a prefix $s = (n, k)$, let

$$\ell_s^{\pm}(d) = (1/2 \pm d)^k (1/2 \mp d)^{n-k}.$$

The ordinary stopping reward is $r_s = \mathfrak{r}(p_H(s) - p_L(s), p_v(s))$. Define the latent-reveal reward

$$g_s = \int \left\{ \frac{\delta(d)^2 \ell_s^+(d)}{q_v(d)} + \frac{\delta(d)^2 \ell_s^-(d)}{1 - q_v(d)} \right\} \mu(dd). \tag{100}$$

Terms with zero numerator and denominator are zero; a nonzero numerator over zero has value infinity. In normalized count coordinates, multiply $r_s, g_s, p_w(s)$ by $\binom{n}{k}$ and use the corresponding Pascal factors.

Let K_N^+ be the optimal information/work ratio in the experiment which uses ordinary observations until it stops before depth N, but receives (D, S) at no extra cost if it reaches depth N. Its Bellman recursion is exactly (88), with terminal value g_{Nk} in place of r_{Nk}. The root condition is unchanged. Thus $\kappa \geq K_N^+$ exactly when this modified recursion has root surplus at most zero.

For a finite latent divergence, set

$$\begin{aligned} C_v &= \int \frac{\delta(d)^2}{q_v(d)(1 - q_v(d))} \, \mu(dd), \\ e_N^v &= C_v - \sum_{k=0}^{N} \binom{N}{k} \mathfrak{r}(p_H(N,k) - p_L(N,k), p_v(N,k)). \end{aligned} \tag{101}$$

Theorem G.2 (Monotone stopping-ratio brackets)**.** *For every admissible quadruple of priors, with information ratios interpreted in the extended nonnegative reals,*

$$K_N \uparrow K_\infty \leq K_N^+, \qquad K_{N+1}^+ \leq K_N^+. \tag{102}$$

If, in addition, $C_v < \infty$, then

$$0 \leq K_N^+ - K_N \leq e_N^v, \qquad e_N^v \longrightarrow 0. \tag{103}$$

Consequently $K_N^+ \downarrow K_\infty$ under this finite-divergence hypothesis. Without it, the monotone brackets remain valid, but pointwise convergence of the upper bracket need not hold.

Proof. Under the variance-reference latent law, define the score

$$Z(D, S) = \begin{cases} \delta(D)/q_v(D), & S = 1, \\ -\delta(D)/(1 - q_v(D)), & S = 0. \end{cases}$$

If $C_v < \infty$, it is square integrable, has mean zero, and has second moment C_v. The score of any observed transcript is its conditional expectation. Thus its signed-likelihood information is $\mathbb{E}_v[\mathbb{E}_v(Z \mid \text{transcript})^2]$.

Take any capped ordinary policy with cap larger than N and stop its paid observations at $T \wedge N$. On reaching N, the genie reveals (D, S); these values allow simulation of every future Bernoulli observation and therefore of the original terminal transcript. Data processing cannot decrease information when passing to the stronger genie transcript, while the work under q_w can only decrease. This proves $K_\infty \leq K_N^+$ by taking the supremum over finite ordinary caps. The same simulation shows that an earlier reveal is stronger than a later reveal, so K_N^+ is nonincreasing. The other monotonicity and $K_N \leq K_\infty$ are immediate. These arguments also apply to infinite divergences using the extended-value definition.

For a cap-N genie policy, erase the reveal. The resulting ordinary policy has exactly the same paid work. Its lost information is

$$\mathbb{E}_v\left[\mathbf{1}_{\{T=N\}}\operatorname{Var}_v(Z \mid \mathcal{F}_N)\right] \le \mathbb{E}_v \operatorname{Var}_v(Z \mid \mathcal{F}_N) = e_N^v.$$

Here $\mathcal{F}_N$ is the first N observations of the completed counterfactual stream. An independent policy seed can be included; it adds no latent information. Since root-active work under q_w is at least one, dividing by work and taking suprema gives $K_N^+ \le K_N + e_N^v$.

The infinite Bernoulli stream identifies its mean almost surely and hence identifies (D, S) because $D > 0$. The martingale $\mathbb{E}_v(Z \mid \mathcal{F}_N)$ therefore converges to Z in L^2. This proves $e_N^v \to 0$. Its explicit formula is precisely (101). □

Remark G.3 (Pointwise convergence can fail at infinite latent divergence). Let $\mu(dd) = 2\,dd$ on $(0, 1/2]$, $\rho = 1/4$, and take $q_L = 0$, $q_H = q_w = 1/4$, and $q_v(d) = 1/2 - d$. All four priors are admissible. Here

$$C_v = \frac{1}{8}\int_0^{1/2} \frac{dd}{1/4 - d^2} = \infty.$$

The policy which always reaches depth N and accepts the latent reveal has infinite information and work N, so $K_N^+ = \infty$ for every finite N.

Nevertheless $K_\infty \le 3/4$. To verify this for every capped policy, parameterize the latent state by its Bernoulli mean p. The variance reference has density $2(1-p)$ on $(0,1)$, the work reference has density $3/2$ below $1/2$ and $1/2$ above $1/2$, and the signed endpoint difference has density $-1/2$ below $1/2$ and $1/2$ above $1/2$. For an ordered prefix s of length $n \ge 1$ with k heads, put $u_s = \int_0^1 p^k(1-p)^{n-k}\,dp$. Then

$$|p_H(s) - p_L(s)| \le u_s/2, \qquad p_v(s) = \frac{2(n-k+1)}{n+2}u_s,$$

and hence

$$r_s \le \frac{n+2}{8(n-k+1)}u_s \le \frac{3n}{8}u_s.$$

Sum over stopped prefixes and the independent seed. If U denotes the auxiliary uniform law of p, the information of any capped policy is at most $(3/8)\mathbb{E}_U T$. The work-reference density dominates half the uniform density, so this is at most $(3/4)\mathbb{E}_{q_w} T$. Taking suprema gives the claimed bound. The auxiliary law U need not be admissible; it is used only for this domination calculation.

This example does not contradict the optimized convergence in the next theorem, whose proof first regularizes the variance reference.

G.3 A convergent lower program and a stopping rule

Define the strengthened-experiment dual value

$$G_N(B)^2 = \sup_{q_L, q_H, q_v, q_w \in \mathcal{Q}_\rho} \frac{[\theta(q_H) - \theta(q_L)]_+^2}{4 + BK_N^+(q_L, q_H; q_v, q_w)}. \tag{104}$$

As before, a ratio with infinite denominator is zero.

Theorem G.4 (A posteriori minimax brackets). *For every law, rarity and budget in the model,*

$$0 < G_N(B) \uparrow P_\infty(B), \qquad P_N(B) \downarrow P_\infty(B), \tag{105}$$

and at every integer $N \geq 1$,

$$\boxed{G_N(B)/8 \leq R_{\mu,\rho}(B) \leq P_N(B).} \tag{106}$$

For finite-support laws both programs have finite conic formulations. The condition $P_N(B) \leq 2G_N(B)$ eventually holds and certifies $R(B) \leq P_N(B) \leq 16R(B)$.

Proof. The preceding theorem gives $G_N \leq P_\infty$ and monotonicity. The constant priors $q_L = 0, q_H = q_v = \rho$ have finite latent divergence and give a strictly positive lower bound for every G_N.

To prove convergence, fix a quadruple with finite K_∞ and positive target gap. Put $m = (q_L + q_H)/2$ and regularize only the variance reference:

$$q_v^\epsilon = (1-\epsilon)q_v + \epsilon m, \qquad 0 < \epsilon < 1.$$

The new transcript reference dominates $(1-\epsilon)$ times the old one, so

$$K_\infty(q_L, q_H; q_v^\epsilon, q_w) \leq \frac{K_\infty(q_L, q_H; q_v, q_w)}{1-\epsilon}.$$

Also $|\delta| \leq 2m$ and $|\delta| \leq 2(1-m)$, so the new latent score has absolute value at most $2/\epsilon$. Its latent divergence is finite. Theorem G.2 now makes its genie ratio converge to its ordinary infinite-cap ratio. Keep $\epsilon > 0$ fixed throughout this limit in N; no convergence of the unregularized quadruple is used. Hence

$$\lim_N G_N(B)^2 \geq \frac{(\Delta\theta)^2}{4 + BK_\infty/(1-\epsilon)}.$$

Let $\epsilon \downarrow 0$ and maximize using Lemma G.1. The limits are taken in this order; no convergence uniform in ϵ is asserted. This proves $G_N \uparrow P_\infty$. Equation (99) gives the risk bracket. Since $P_\infty > 0$, the ratio P_N/G_N tends to one. The stated stopping criterion therefore succeeds at a finite cap, with no claimed uniform rate for that cap. □

For an explicitly given law $\mu = \sum_i w_i \delta_{d_i}$, the genie dual is a finite SOCP. In the price formulation, use endpoint variables $u_H = \alpha q_H$, $u_L = \alpha q_L$, work-reference variables $e = \eta q_w$, and a variance reference q_v. Their rarity and box constraints are linear perspective constraints. For each atom introduce

$$(u_{H,i} - u_{L,i})^2 \leq q_{v,i} r_i^+, \qquad (u_{H,i} - u_{L,i})^2 \leq (1 - q_{v,i}) r_i^-.$$

At terminal states replace the ordinary quadratic-over-linear stopping cone by the linear bound

$$X_s \geq \frac{1}{4} \sum_i w_i \{\ell_s^+(d_i) r_i^+ + \ell_s^-(d_i) r_i^-\}.$$

Keep the ordinary stopping cones at earlier states and the same linear continuation/root inequalities. Maximize $\sum_i w_i(u_{H,i} - u_{L,i}) - \alpha^2 - \eta B$. This is exactly the squared value in (104) after eliminating the price and scale. It uses $O(N^2 + m)$ variables, apart from the coefficient matrices. For a general measure the result remains a variational statement; a finite-support computational representation is not silently assumed for an arbitrary law.

G.4 Certificates that can be checked independently

In practice a feasible primal policy gives an upper bound $U \geq R$, and a feasible genie dual gives a square-root lower bound $L \leq G_N$. Consequently the *verified* inequality $U \leq 2L$ is sufficient for a factor-16 risk certificate. An approximate solver status or an uncertified objective is not sufficient. Section I uses exact rational checks for the finite rational examples.

A selected dual quadruple can also yield a stronger direct risk lower bound than $L/8$. Absorb its two references into its endpoints with equal thirds, and let t be the new target separation. If a Bellman certificate gives $\kappa \geq K_N^+$ for the original quadruple, the absorbed midpoint pair has $K_\infty \leq \kappa$. With $x = B\kappa$, affinity testing gives

$$R(B) \geq \max_{0\leq s\leq 1} \frac{st}{2}\left(1 - \frac{s^2 x}{4}\right)_+. \tag{107}$$

The square of this bound is the rational expression

$$\begin{cases} \dfrac{t^2}{4}(1-x/4)^2, & 0 \leq x \leq 4/3, \\ \dfrac{4t^2}{27x}, & x \geq 4/3. \end{cases} \tag{108}$$

This supplies an exact squared lower certificate when the inputs and the certified price are rational.

H What a fixed observation cap can identify

The exact dual also identifies an obstruction that is invisible in a small-cap numerical upper bound: increasing the number of units cannot resolve two priors with identical cap-N one-unit distributions.

Let $L = \lceil N/2 \rceil$ and define the moment-matching diameter

$$\Delta_N = \max\left\{\theta(q_H) - \theta(q_L) : q_L, q_H \in \mathcal{Q}_\rho,\ \int (q_H - q_L) d^{2j+1}\, d\mu = 0\ (0 \leq j < L)\right\}. \tag{109}$$

Compactness ensures that this maximum is attained.

Theorem H.1 (Exact fixed-cap bias floor)**.** *For every magnitude law,*

$$\lim_{B\to\infty} P_N(B) = \Delta_N/2. \tag{110}$$

If μ has exactly m distinct positive support points, then $\Delta_N = 0$ if and only if $N \geq 2m-1$. If μ has infinite support, then $\Delta_N > 0$ for every finite N.

Proof. The distribution of N Bernoulli observations from one unit determines the first N moments of its latent Bernoulli mean. The difference of the kth such moments under two priors is

$$\int (q_H - q_L)\{(1/2+d)^k - (1/2-d)^k\}\, d\mu.$$

These polynomials span exactly the odd monomials $d, d^3, \ldots, d^{2L-1}$. Thus the constraints in (109) are equivalent to equality of the full depth-N count distributions. Equality at depth N implies equality for every stopped experiment with that cap, by marginalization and independent randomization. Conversely, the policy that always observes exactly N samples witnesses a positive information numerator whenever those count distributions differ. Therefore $K_N = 0$ exactly for pairs satisfying (109); otherwise $K_N > 0$ or $K_N = \infty$.

Apply the decreasing compact-set argument from Lemma G.1 to the dual ratios as $B \to \infty$. Their pointwise limit is $(\Delta\theta)_+^2/4$ when $K_N = 0$, and zero otherwise. Theorem E.1 gives (110).

For the support assertion, consider the finite-dimensional subspace

$$\mathcal{V}_L = \operatorname{span}\{d, d^3, \ldots, d^{2L-1}\} \subset L^2(\mu).$$

If the support has $m \le L$ points, polynomial interpolation gives $p(d) = dQ(d^2) = 1$ on that support, with $\deg Q \le m-1$. Then $1 \in \mathcal{V}_L$, so moment matching forces zero target gap.

If the support has more than L points, $1 \notin \mathcal{V}_L$. Indeed, a polynomial Q of degree at most $L-1$ cannot agree with $x^{-1/2}$ at $L+1$ distinct positive values: otherwise repeated Rolle's theorem would force the nonvanishing Lth derivative of $x^{-1/2}$ to vanish somewhere. The same argument covers infinite support, since an almost-everywhere polynomial identity extends to the support.

Let $h = 1 - \mathrm{proj}_{\mathcal{V}_L} 1$. It is a nonzero bounded function, orthogonal to every required odd monomial, and

$$\int h \, d\mu = \int h^2 \, d\mu > 0.$$

Set $\epsilon = \rho/(4\|h\|_\infty)$ and $q_H = \rho/2 + \epsilon h$, $q_L = \rho/2 - \epsilon h$. Both priors lie pointwise between $\rho/4$ and $3\rho/4$ and hence belong to $\mathcal{Q}_\rho$. They match the odd moments but have positive target gap $2\epsilon \int h^2 \, d\mu$. This proves strict positivity. Finally $L \ge m$ is equivalent to $N \ge 2m-1$. □

The proof also yields the explicit lower estimate

$$\Delta_N/2 \ge \frac{\rho \|h\|^2_{L^2(\mu)}}{4\|h\|_{L^\infty(\mu)}} \quad \text{when } 1 \notin \mathcal{V}_L.$$

This theorem identifies the target functional from cap-limited laws; it does not require recovery of the entire allocation q. This theorem concerns the fixed-cap program and its infinite-budget limit. It does not impose a bias floor on the unrestricted experiment. The monotone brackets in Section G increase the cap when needed, and their stopping criterion tests the remaining gap directly.

I Exact numerical certificates on finite laws

The computations in this section produce two-sided certificates for eight finite rational instances. Numerical conic optimization selects candidate policies and dual priors. Separate rational-arithmetic verifiers then check the resulting bounds, including the observation budget and the stopping criterion. An approximate solver objective is never used as a certified endpoint.

I.1 What the two endpoints mean

Write U_N for the risk bound of a verified observable cap-N policy, and L_N for the value of a verified feasible genie dual. Their roles are

$$L_N \le G_N \le P_\infty \le P_N \le U_N, \qquad L_N/8 \le R_{\mu,\rho}(B) \le U_N.$$

Thus L_N is a lower bound on a program value. It is not, by itself, the reported lower bound on fully adaptive minimax risk. We also compute the stronger pair-specific adaptive-risk lower bound $\underline{r}_N$ from (107)–(108). Its target separation is that of the absorbed midpoint pair.

The cap search stops only after the exact rational comparison

$$U_N^2 \le 4L_N^2 \tag{111}$$

has passed. This implies $U_N \le 16 R_{\mu,\rho}(B)$. The comparison uses the stored rational squared values, not the rounded decimals in the tables.

I.2 The benchmark laws and results

Every instance has $\rho = 1/10$. The four laws are

$$\begin{aligned}\mu_{\text{one}} &= \delta_{1/4},\\ \mu_{\text{three}} &= \tfrac{3}{4}\delta_{2/5} + \tfrac{1}{5}\delta_{1/5} + \tfrac{1}{20}\delta_{2/25},\\ \mu_{\text{hard}} &= \tfrac{99}{100}\delta_{9/20} + \tfrac{1}{100}\delta_{2/25},\\ \mu_{\text{label}} &= \delta_{1/2}.\end{aligned}$$

Each was run at $B = 100$ and $B = 1000$. The candidate caps were $4, 8, 16, 32$, with the perfect-label cases starting at $N = 1$. Seven instances passed (111) at their first cap. The three-scale law at $B = 1000$ failed at $N = 4$ and passed at $N = 8$. There were therefore nine attempted brackets and eight successful ones. The failed bracket is retained in the source bundle; it remains a valid bracket but does not meet the requested factor-two program criterion.

Law	B	N	L_N	$\underline{r}_N$	U_N
One gap	100	4	0.03474453	0.00619890	0.03891866
One gap	1000	4	0.01461177	0.00196026	0.01825093
Three scales	100	4	0.03331182	0.00573109	0.04551457
Three scales	1000	8	0.01569090	0.00212025	0.02553208
Rare hard atom	100	4	0.02912416	0.00459699	0.03009854
Rare hard atom	1000	4	0.01105026	0.00145369	0.01166586
Perfect labels	100	1	0.02572478	0.00384900	0.02572479
Perfect labels	1000	1	0.00932054	0.00121716	0.00932055

Table 1: Successful exact brackets. Both lower columns are rounded downward and the upper column upward. The adaptive minimax risk lies in $[\underline{r}_N, U_N]$; the unrestricted flow-program value lies in $[L_N, U_N]$. These two intervals have different meanings.

For the failed three-scale attempt at $B = 1000, N = 4$, the verified program endpoints, rounded outward, are 0.01359364 and 0.02989647, and the exact ratio exceeds two. At $N = 8$ the program ratio is about 1.6272, so the exact test passes. Across the eight successful instances, the program bracket ratios range from essentially one to 1.6272. The sharper adaptive-risk bracket ratios range from about 6.27 to 12.05, all below sixteen.

These are small benchmark instances. Their modest successful caps provide no uniform bound on the cap or runtime needed for another law, rarity, or budget. The convergence theorem is qualitative and the certificates concern the particular rational inputs displayed above.

I.3 An analytic check with perfect labels

For $\mu = \delta_{1/2}$, the first observation reveals the sign exactly. Further observations add no information. For every cap $N \geq 1$,

$$P_N(B)^2 = P_\infty(B)^2 = G_N(B)^2 = \frac{ac}{a+c}, \qquad a = \frac{\rho^2}{4}, \quad c = \frac{\rho(1-\rho)}{B}. \tag{112}$$

This is an analytic identity for the flow program, without an assertion that it is the exact fully adaptive minimax risk.

To verify (112), conditional averaging removes randomness in the output given the revealed sign and cannot increase its second moment. Let $\lambda \leq B$ be the active Poisson intensity, let y_0, y_1 be the two outputs, and put $s = \lambda(y_1 - y_0)$. Optimizing the deterministic offset gives worst bias

$\rho|1-s|/2$. At the prior $q=\rho$, centering the outputs minimizes the raw second moment, giving the variance lower bound $s^2\rho(1-\rho)/\lambda$. It is attained by

$$y_0 = -\rho s/\lambda, \qquad y_1 = (1-\rho)s/\lambda.$$

Since $\rho \le 1/4$, their second moment is increasing in q on $[0,\rho]$, so its worst value occurs at ρ. Choosing $\lambda = B$ therefore reduces the objective exactly to $a(1-s)^2 + cs^2$, whose minimum is $ac/(a+c)$. The genie already reveals nothing new after the first observation, which proves the same equality for G_N.

For the two stored perfect-label instances, the exact rational analytic value lies between the exact rational lower and upper certificates. This was checked directly, without rounded decimal comparisons.

I.4 How the certificates are repaired and checked

The conic solver may return a status indicating limited numerical accuracy. Such a status neither establishes nor invalidates a certificate. The following separate checks determine feasibility.

For a lower certificate, the selected priors are quantized to rational numbers, clipped to $[0,1]$, and, if needed, scaled to satisfy the rarity constraint exactly. Variance and work references are regularized by mixing a weight 10^{-8} of the constant prior $\rho/2$. All binomial kernels and latent rewards are then recomputed exactly. A numerical bisection proposes the work price. A new backward Bellman pass uses rational arithmetic and rounds each potential upward to a dyadic grid of denominator 2^{96}. The price is increased if the exact root inequality does not hold. An independent verifier reconstructs the kernels and checks every rarity, stopping, continuation, terminal, and root inequality.

For an upper certificate, a separate verifier works with the actual observable policy: its stopping probabilities at each count state, terminal outputs, Poisson intensity, and deterministic offset. It computes exact conditional means, second moments, and observation counts for every atom and sign. Total stopping probability must equal one exactly. The worst-prior linear expressions are evaluated by an exact fractional-knapsack calculation over $\mathcal{Q}_\rho$. If rationalization raises expected work above B, the intensity is reduced by the exact required factor and outputs are increased by its reciprocal. This preserves means, and the variance bound is recomputed. The reported risk upper is

$$U_N^2 = \left(\sup_{q\in\mathcal{Q}_\rho} |\operatorname{Bias}_q| \right)^2 + \sup_{q\in\mathcal{Q}_\rho} \operatorname{Var}_q .$$

The two maxima need not occur at the same prior; their sum remains a valid upper bound. Integer square-root calculations round every displayed lower RMS downward and every displayed upper RMS upward. The verifier also checks the direction of the stored decimal rounding.

In addition to the optimized genie brackets, the bundle includes eight ordinary finite-cap dual checks and sixteen fixed-quartet midpoint-genie checks. Standard-library-only re-verification accepted all 33 stored lower certificates and nine upper policies. Six targeted corruptions were rejected: a rarity violation, an insufficient Bellman price, an overstated lower value, an insufficient genie terminal reward, excessive actual policy work, and an understated upper risk. These checks remain active when Python is invoked with its optimization flag.

I.5 Standalone reproduction

The source directory `checks/` is self-contained. It includes a clearly attributed copy of the unchanged round-three primal solver, the independent dual solver, the exact verifiers, and all

certificate files. It does not import a solver from a neighboring research directory. From `checks/`, verification needs only Python's standard library:

```
python self_check_certificates.py
python verify_finite_dual.py \
  optimized_genie/three_scales_N8_B1000_lower.json
python verify_policy_upper.py \
  optimized_genie/three_scales_N8_B1000_upper.json
```

The numerical search used Python 3.12 with CVXPY 1.9.3, NumPy 2.5.4, SciPy 1.18.1, and Clarabel 0.11.1. These observed versions are pinned in `requirements.txt`. The principal search is reproduced by

```
python -m pip install -r requirements.txt
python solve_genie_bracket.py
```

Re-running a numerical solver may select different feasible priors and slightly different bounds. The mathematical guarantee comes from the independently checked rational artifacts.

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